

BT

PWR

UG3V3

L5700

600ohm@100MHz

0.165ohm DCR @ 500mA

220

202

100n

10p

C5703

C5706

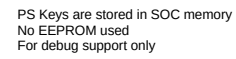
C5700

C5712

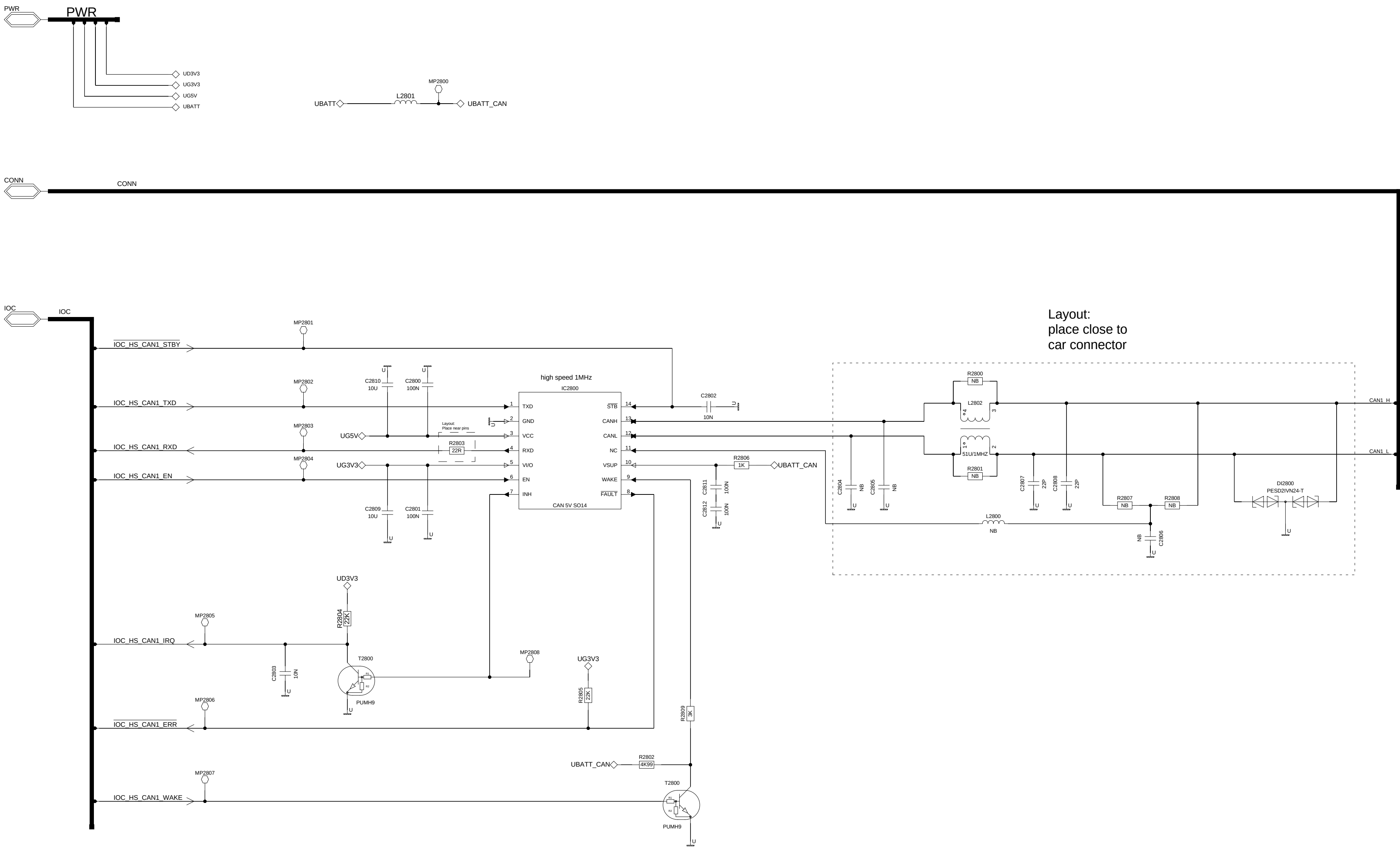
MP5701

BT_VREG_I

100mA Max.

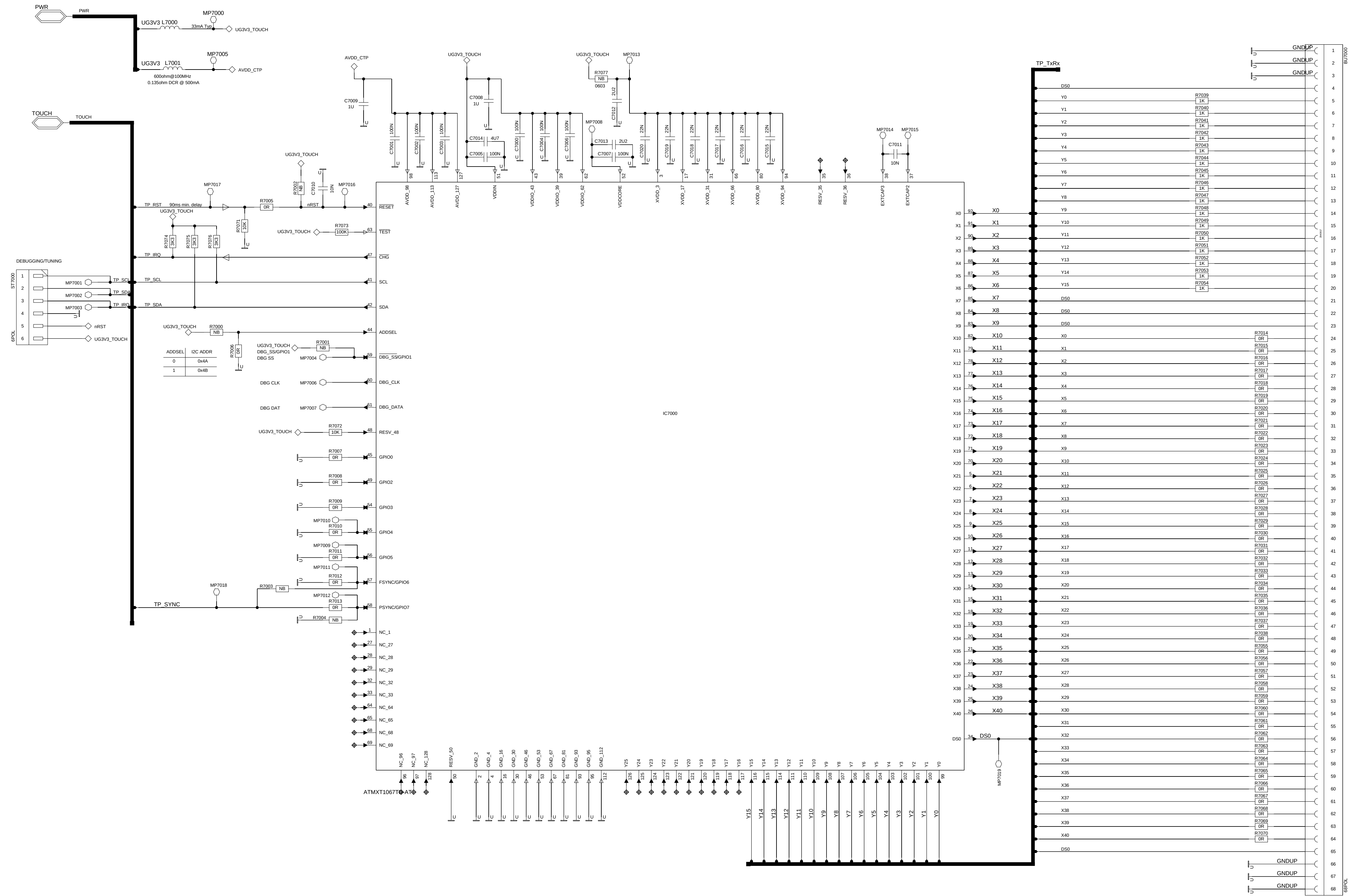
A2 26.02.12

CAN (2800-2899)

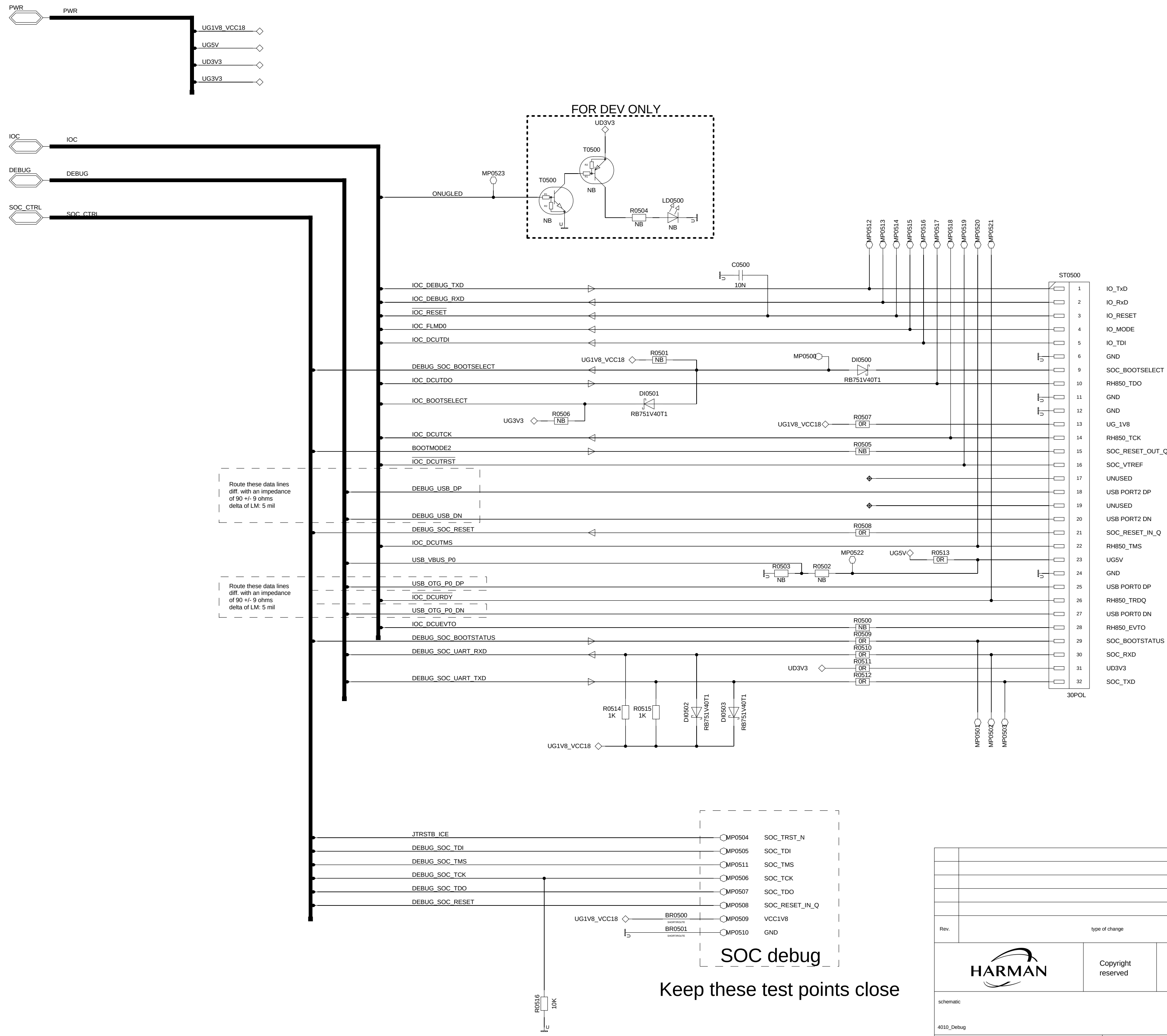


Rev.	type of change		change-no.	date of change	auth. checked				
		Copyright reserved	Wiring modifications reserved			date		name	
					created				
					checked				
schematic				Sheet	Sheets of		Sample		
4010_CAN				3	31				
substitute for:				Type	Variant		Part Number		
substitute by:									

CTP (7000-7099)



Rev.	type of change					change no.	date of change		auth.	checked
 Copyright reserved Writing modifications reserved						date		name		
						created				
						checked				
schematic						Sheet 4	Sheets of 31	Sample		
4010_CTP						Type	Variant C143		Part Number	
substitute for:						substitute by:				



The reproduction, distribution and utilization of this document as well as the communication of its contents to others without explicit authorization is prohibited. Offenders will be held liable for the payment of damages. All rights reserved in the event of the grant of a patent, utility model or design.

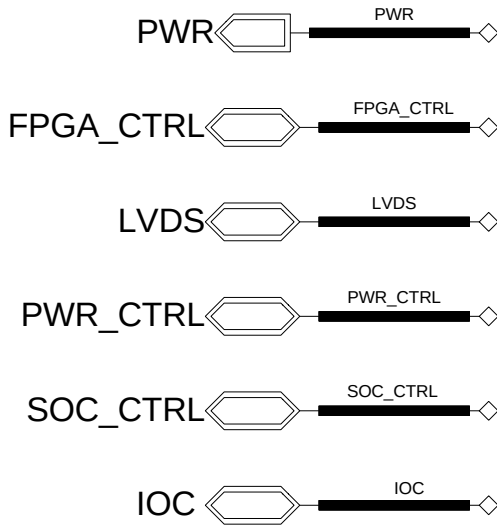
Rev.	type of change		change-no.	date of change	auth. checked
		Copyright reserved	Wiring modifications reserved	date name	
				created	
				checked	
schematic			Sheet 5	Sheets of 31	Sample
4010_Debug			Type	Variant	Part Number
substitute for:		substitute by:			

1) - Route as differential pair signals.
 - Differential impedance: 100ohm (±10%)
 - Maximum intra-pair skew: 0.127mm (5mils)

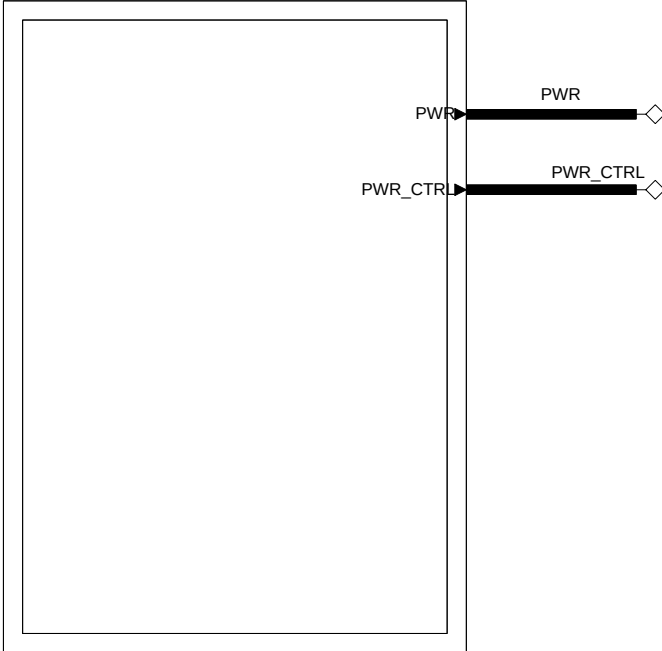
Marvell recommends using Varistors for ESD protection.
 TDK part # AVRH10C101KT1R1NE8

Marvell recommends using Varistors
for ESD protection.
TDK part # AVRH10C101KT1R1NE8

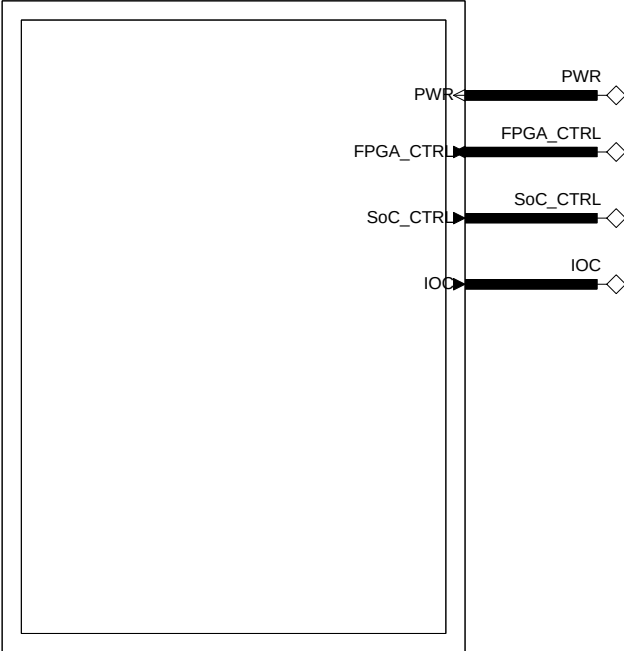
Rev.	type of change		change-no.	date of change	auth. checked
		Copyright reserved			name
			created		
			checked		
schematic			Sheet 6	Sheets of 31	Sample
4010_ETH_PHY			Type	Variant	Part Number
substitute for:		substitute by:			



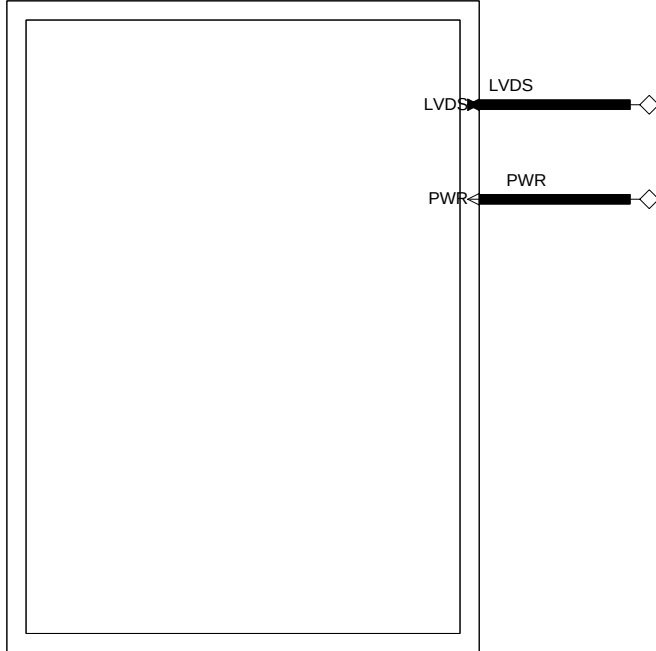
FPGA_POWER



FPGA_CONF

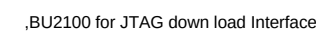
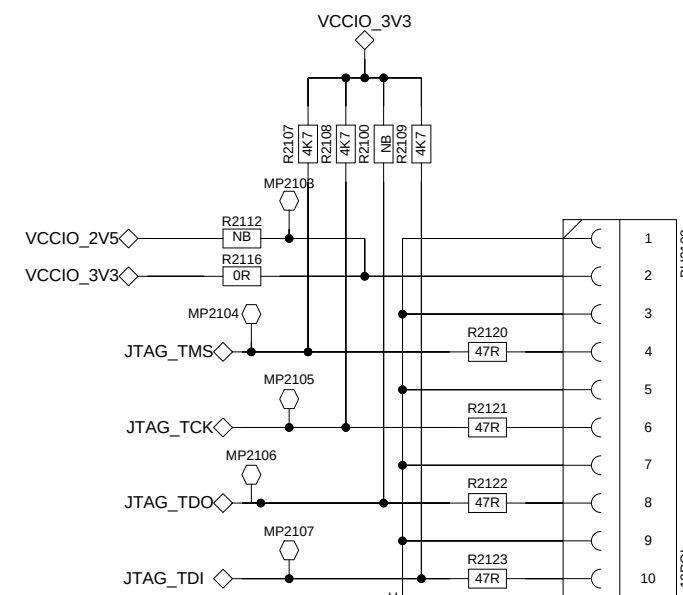
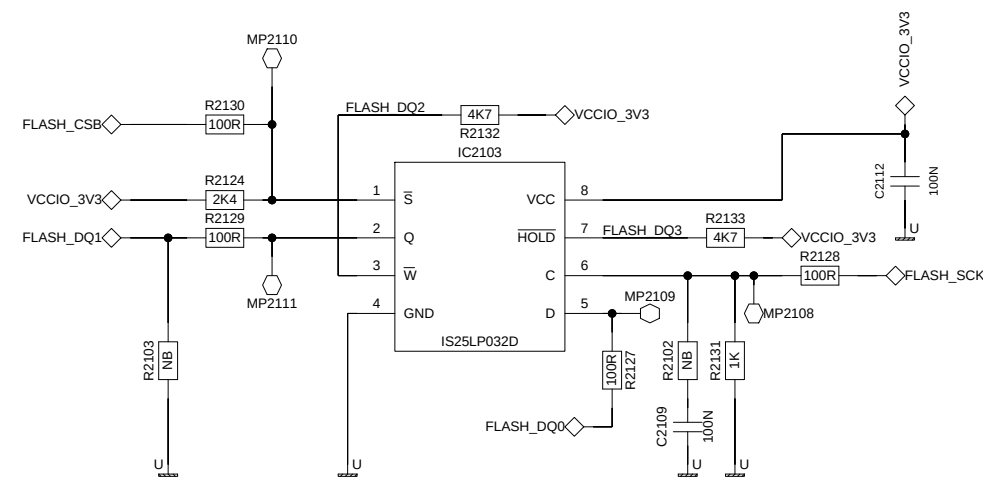
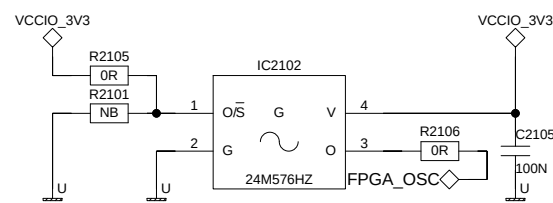
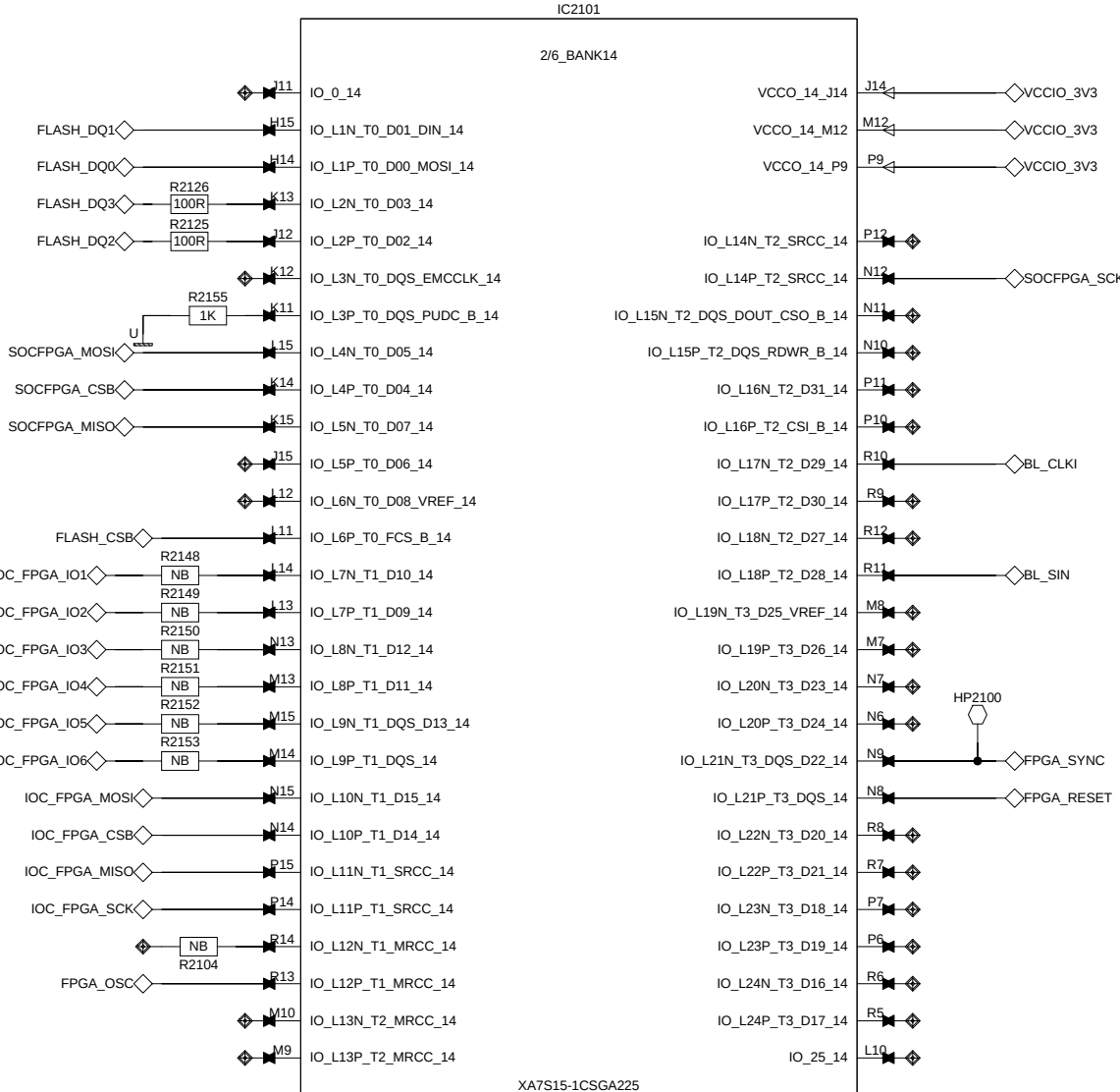
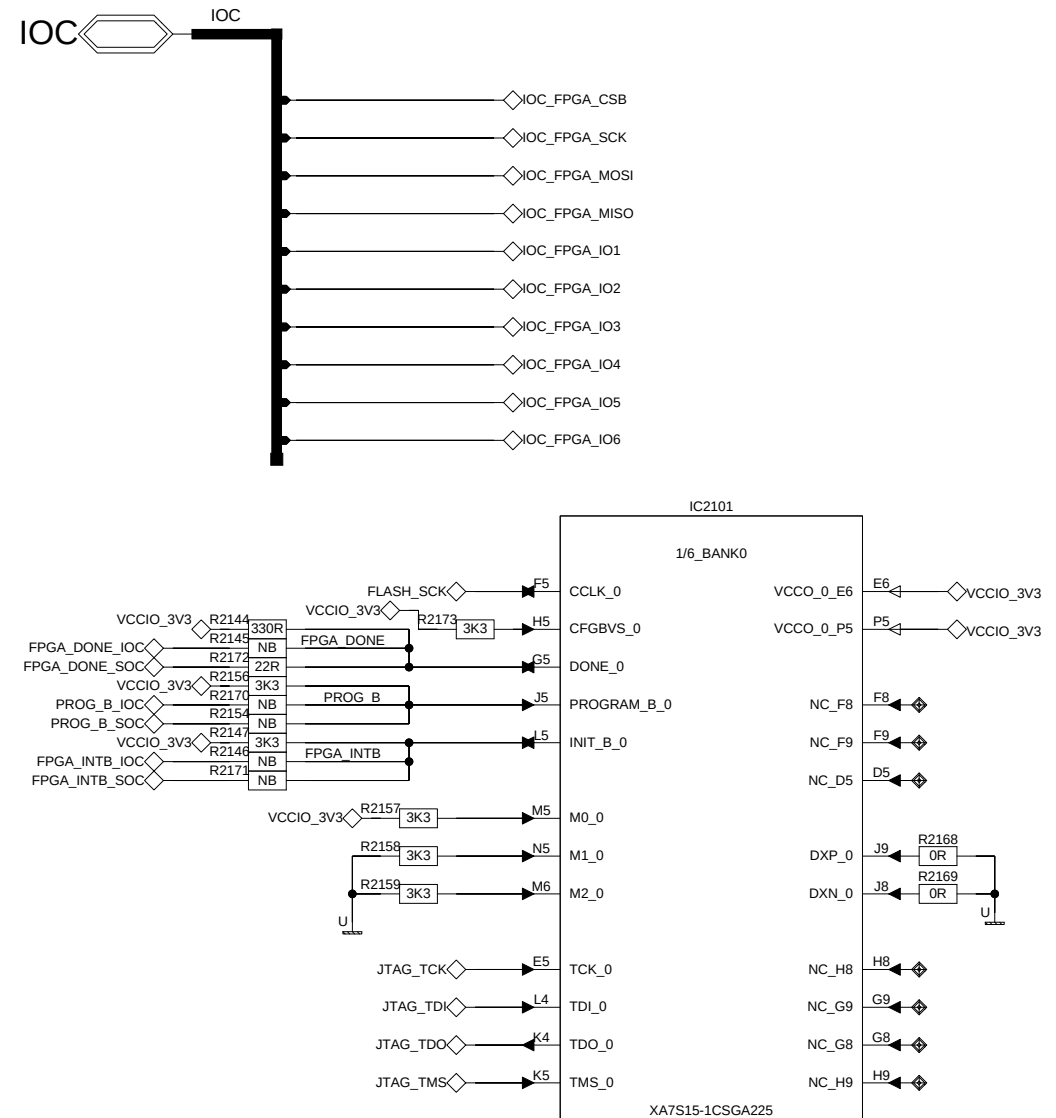
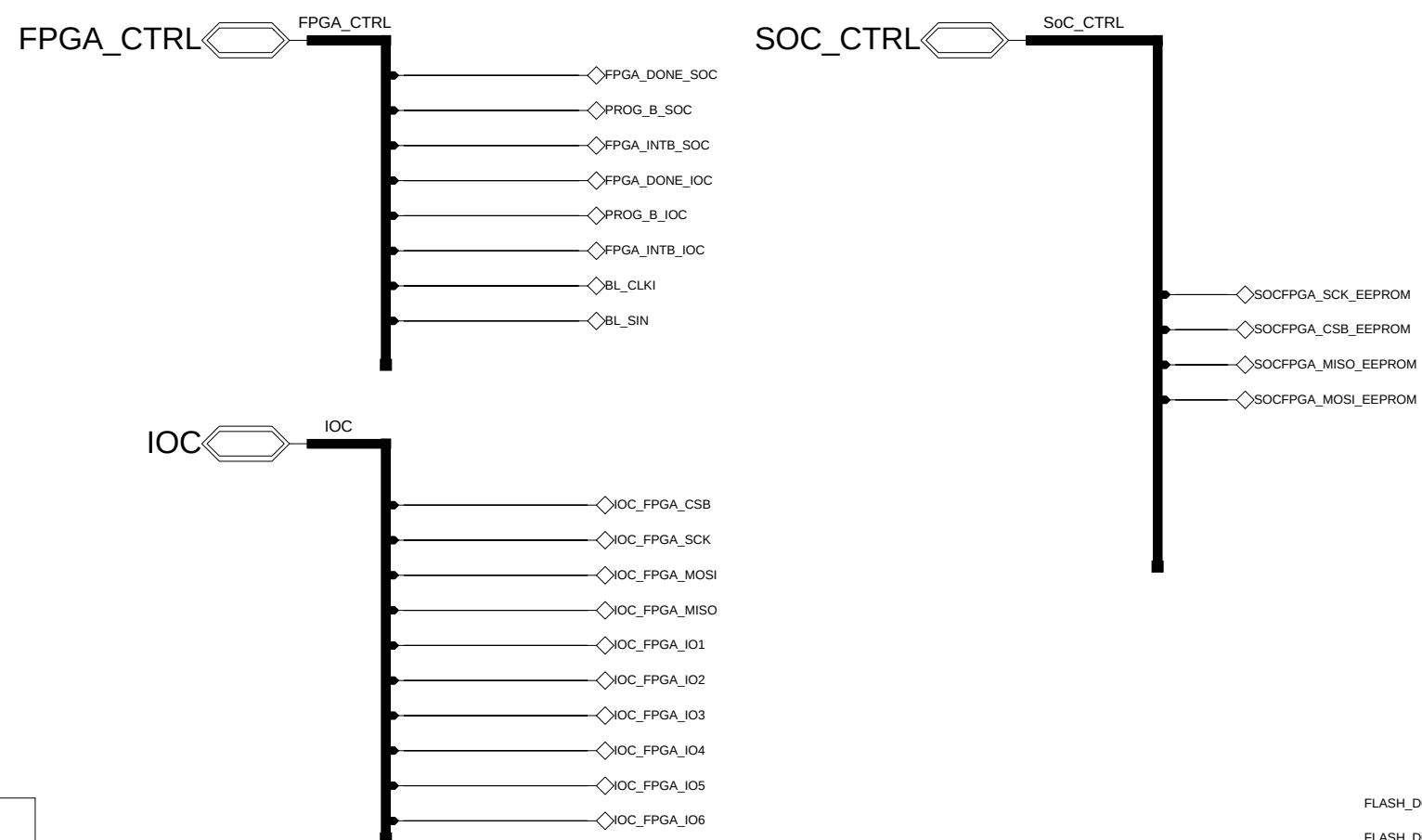


FPGA_VIDEO

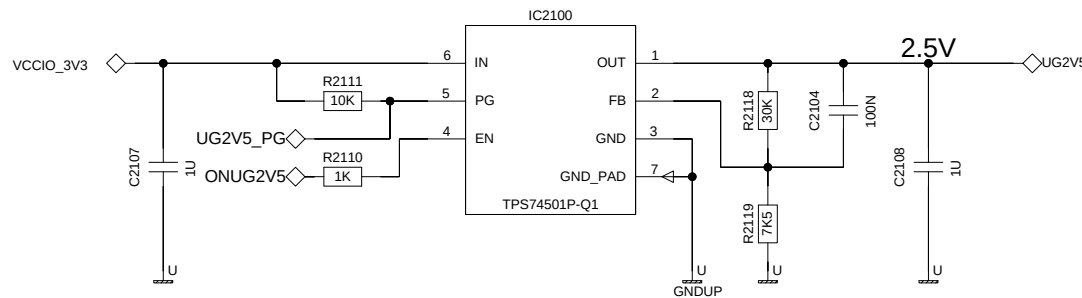
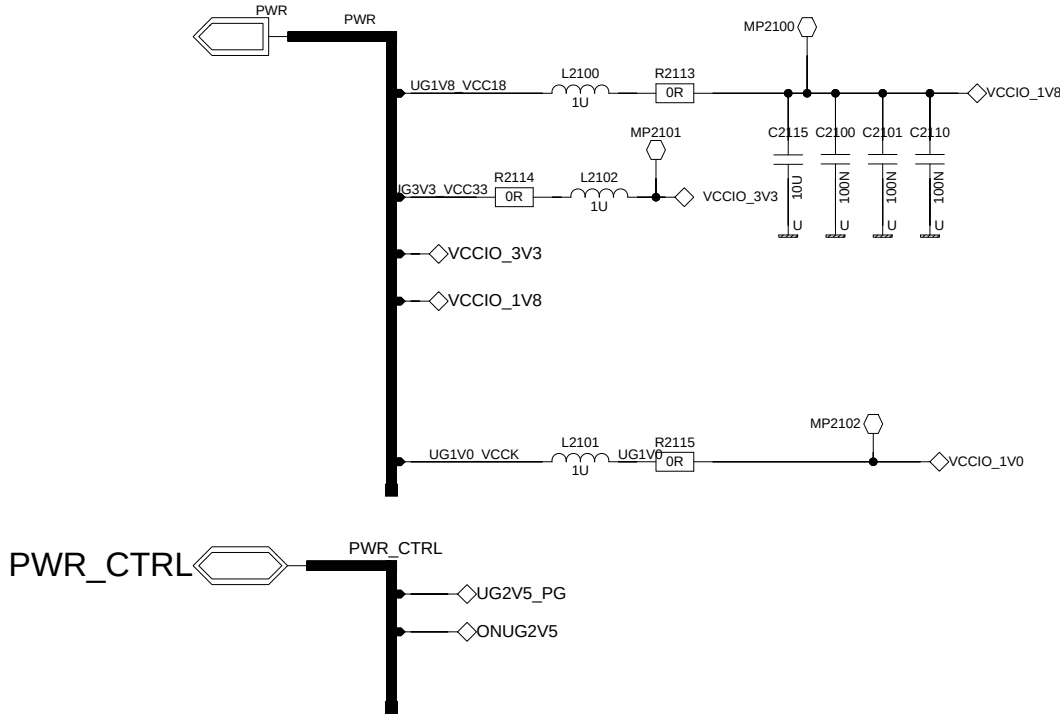
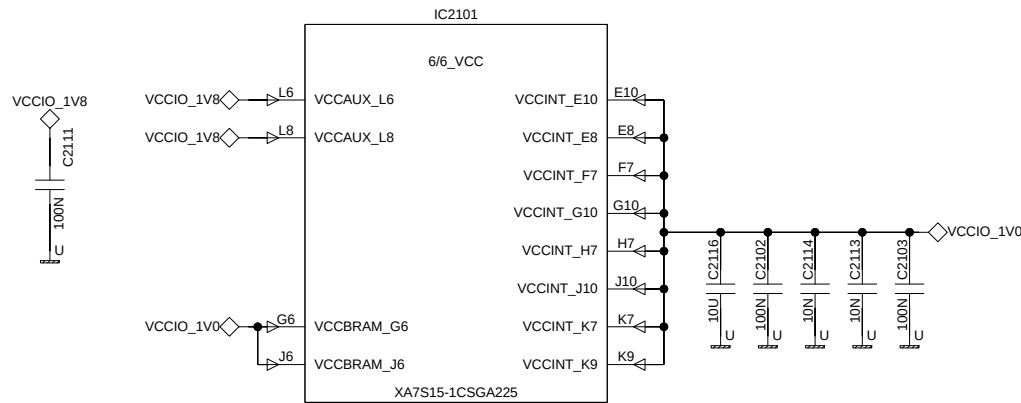
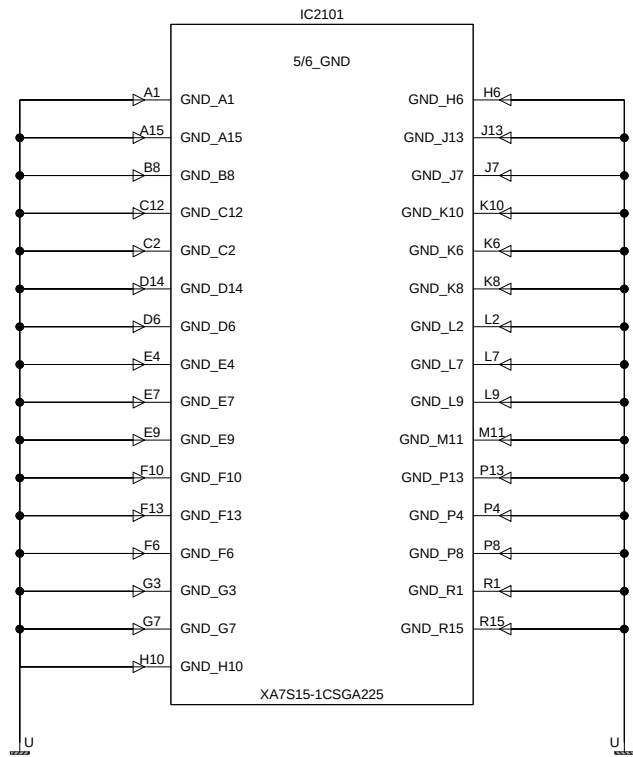


The reproduction, distribution and circulation of this document as well as the communication of its contents to others without explicit authorization is prohibited. Offenders will be held liable for the payment of damages. All rights reserved. A violation of these terms is a criminal offense under the law.

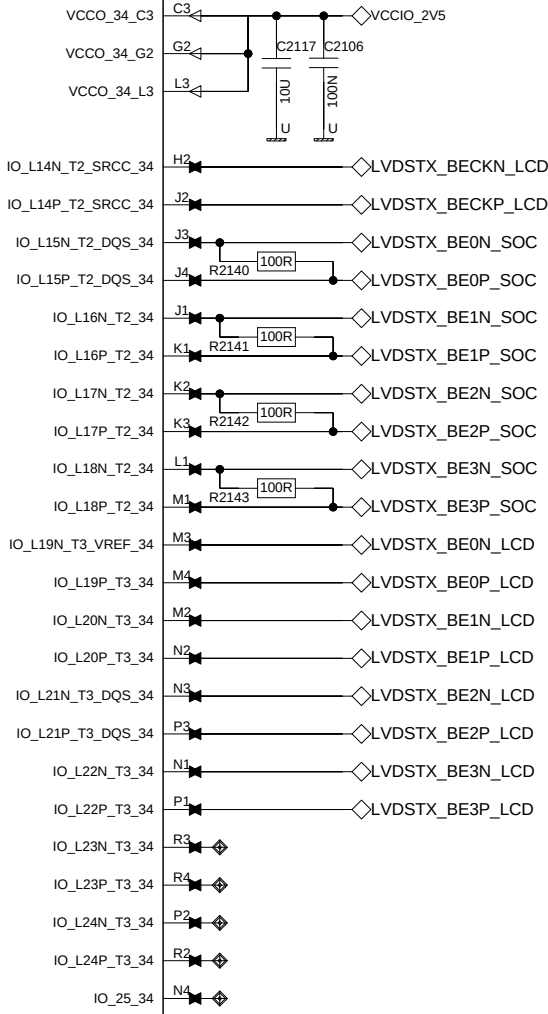
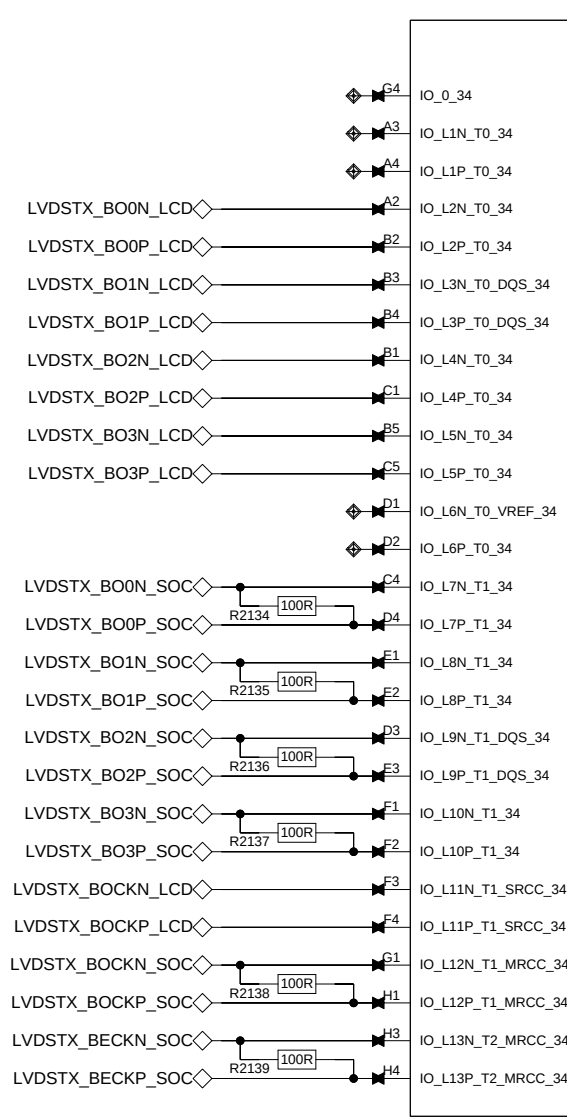
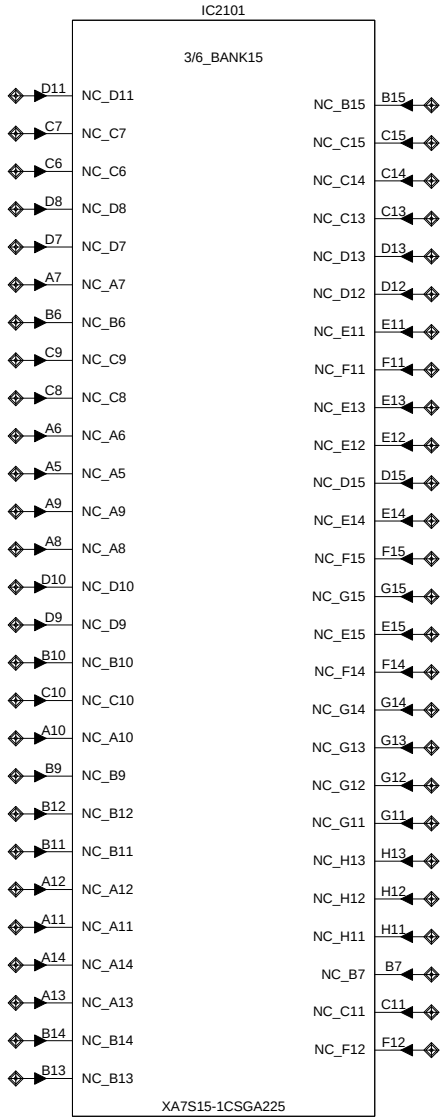
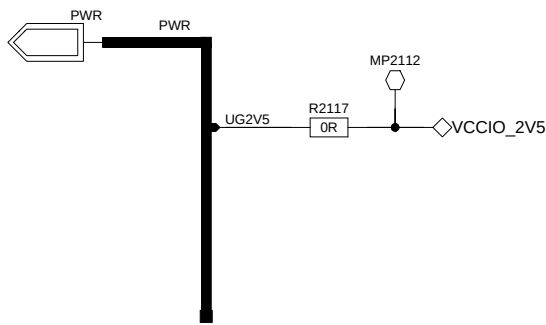
Rev.	type of change				change no.	date of change		auth.	checked
 Copyright reserved Writing modifications reserved					date		name		
					created				
					checked				
schematic:					Sheet 7	Sheets of 31		Sample	
4030_FPGA					Type	Variant C143		Part Number	
substitute for:			substitute by:						



Rev.	type of change		change-no.	date of change	auth. checked
		Copyright reserved	Wiring modifications reserved	date name	
				created	
				checked	
schematic			Sheet 8	Sheets of 31	Sample
FPGA_CONF			Type	Variant	Part Number
substitute for:		substitute by:		C143	

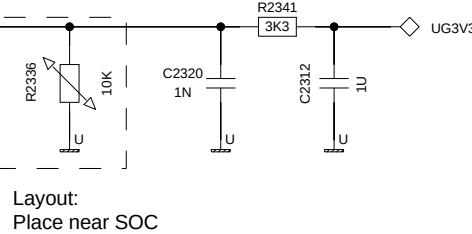


Rev.	type of change	change-no.	date of change	auth.	checked
	Copyright reserved	date		name	
		created			
		checked			
schematic	Sheet		Sheets of		Sample
FPGA_POWER	Type		Variant		Part Number
substitute for:		substitute by:		C143	



Rev.	type of change		change-no.	date of change	auth.	checked
		Copyright reserved	Wiring modifications reserved	date		name
				created		
				checked		
schematic				Sheet	Sheets of	Sample
FPGA_VIDEO				10	31	
substitute for:			Type	Variant	Part Number	
				C143		

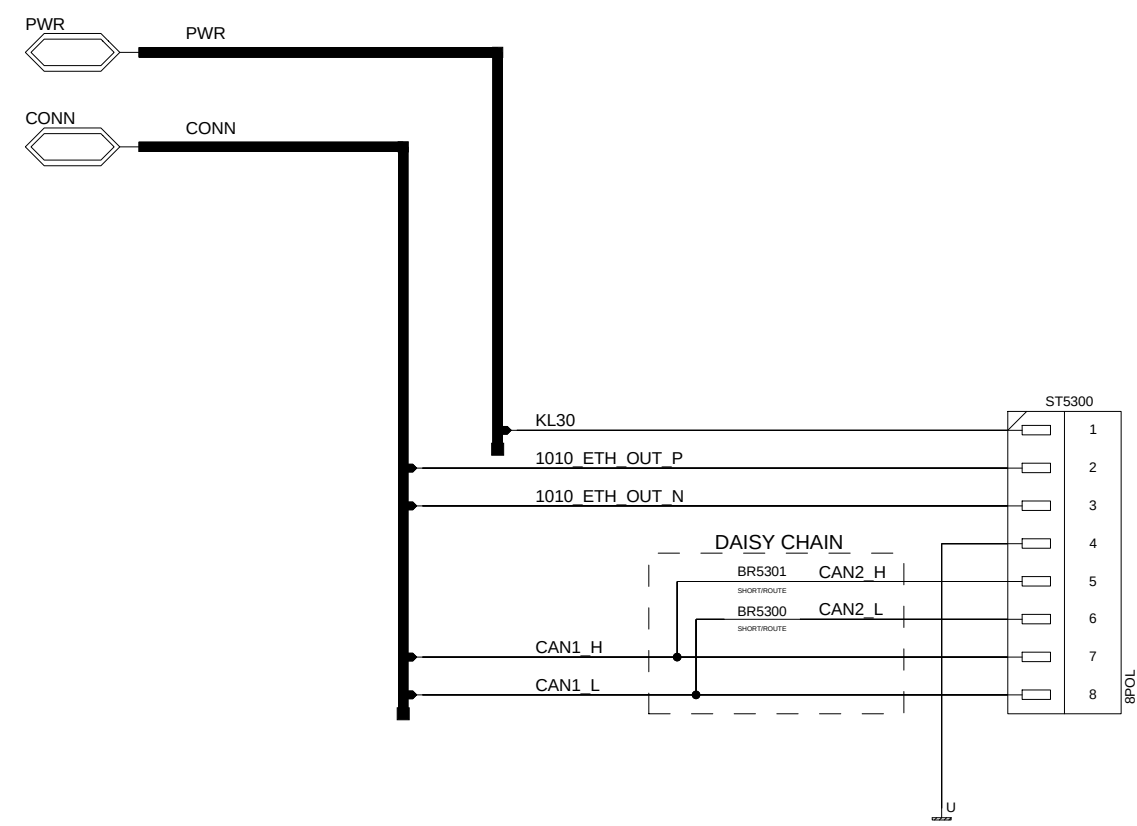
A2 24.02 12



INTP0	AWO	IOC_BKLT_FAULT
INTP1	ISO	-----
INTP2	ISO	-----
INTP3	ISO	HS_CAN1_RXD
INTP4	ISO	-----
INTP5	AWO	-----
INTP6	AWO	-----
INTP7	AWO	-----
INTP8	AWO	HS_CAN1_IRQ
INTP9	AWO	-----
INTP10	ISO	IOC_SOC_IPC_IRQ
INTP11	AWO	-----
INTP12	AWO	-----
INTP13	AWO	-----
INTP14	AWO	DEBUG_RX
INTP15	AWO	-----

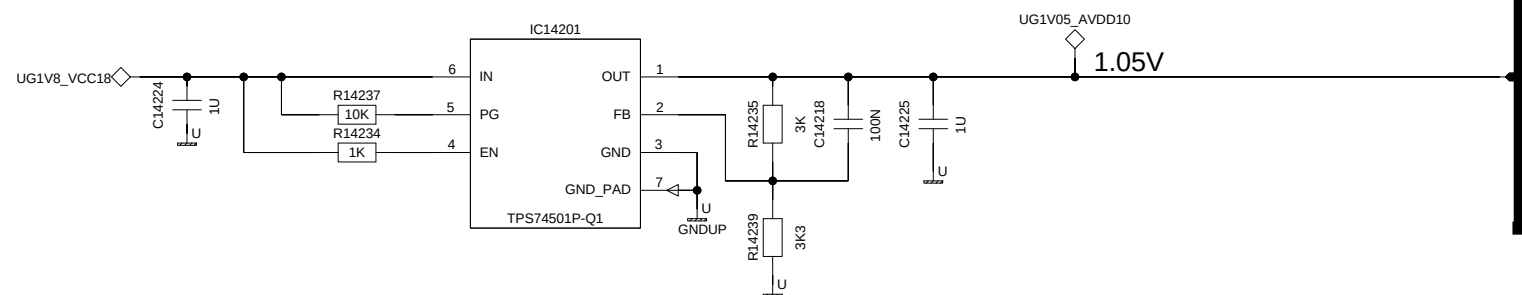
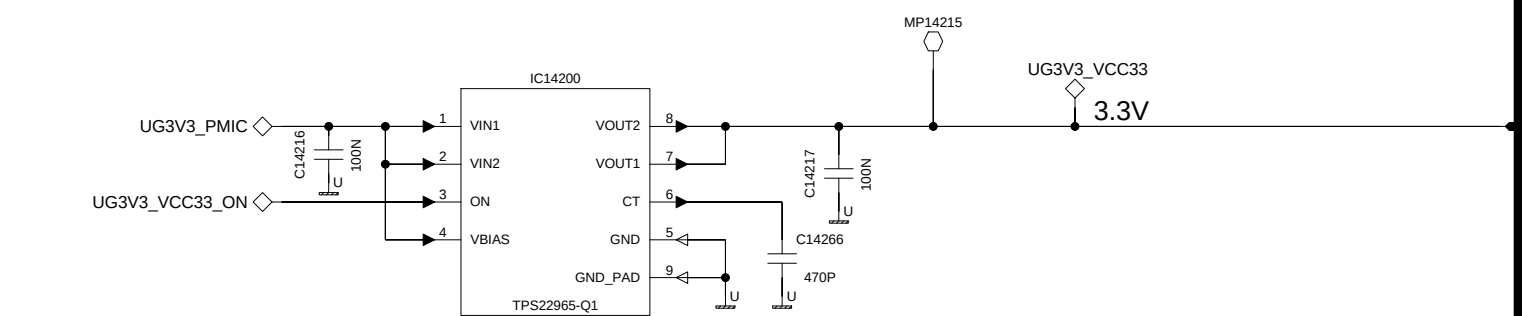
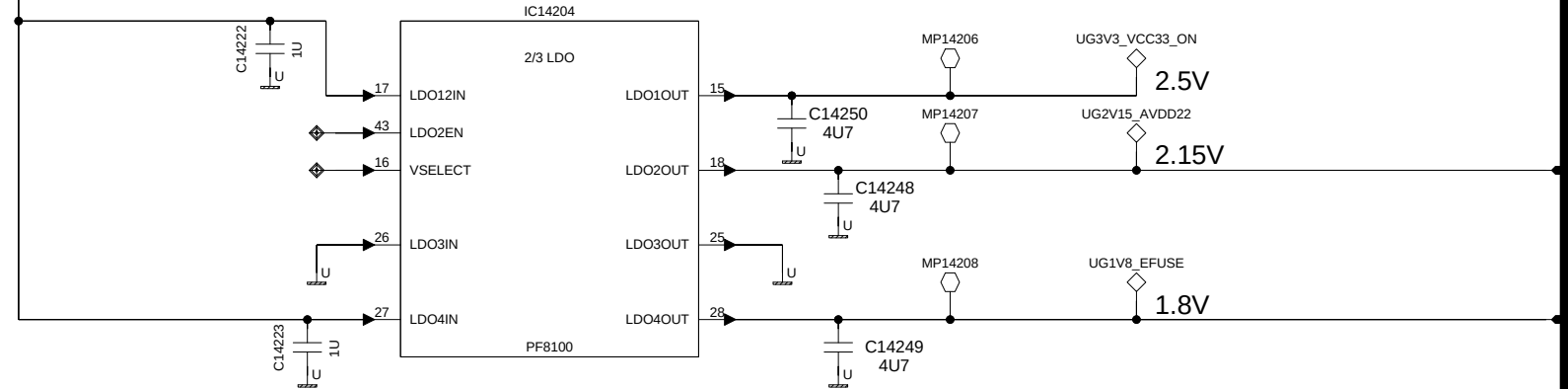
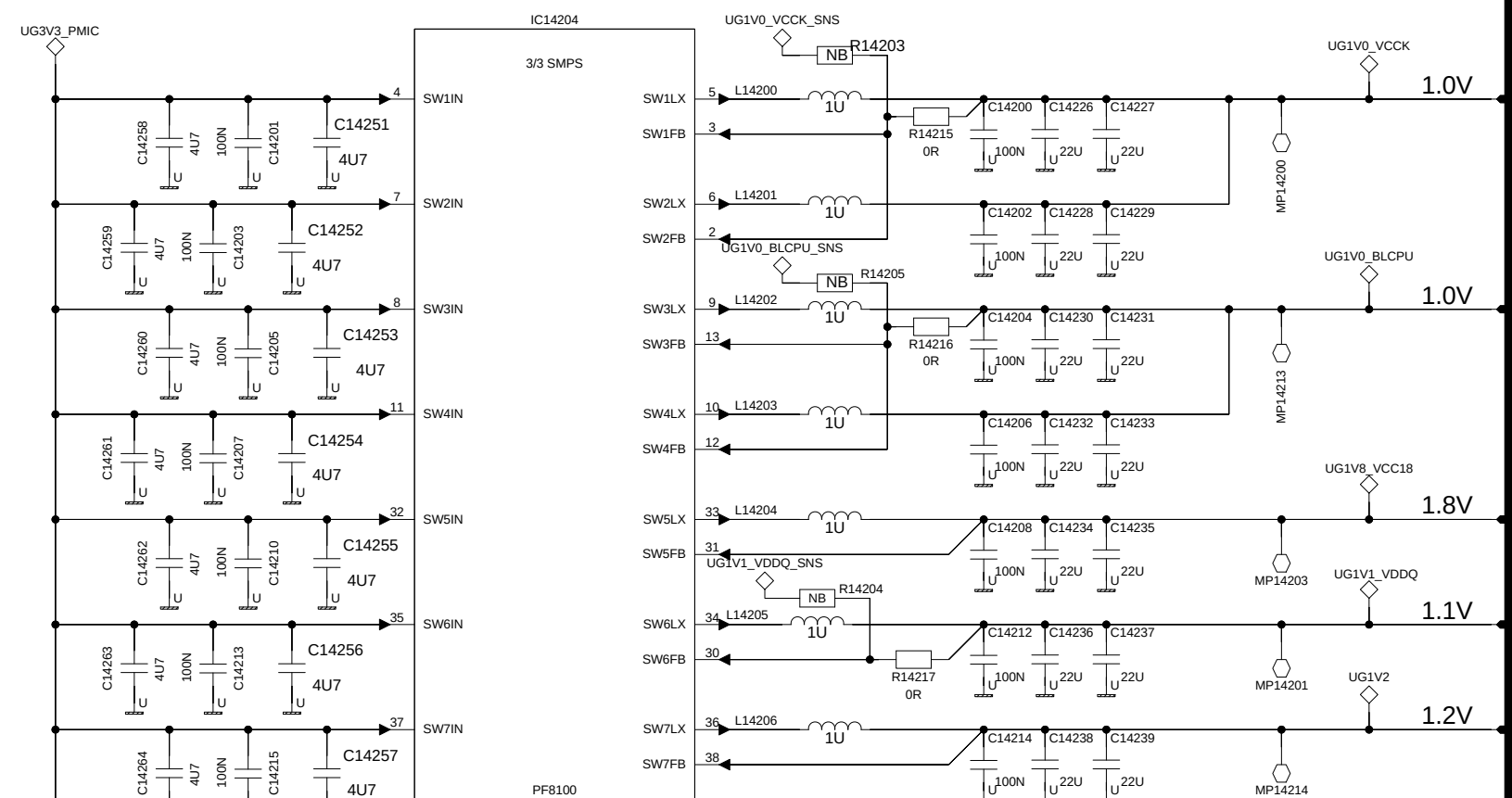
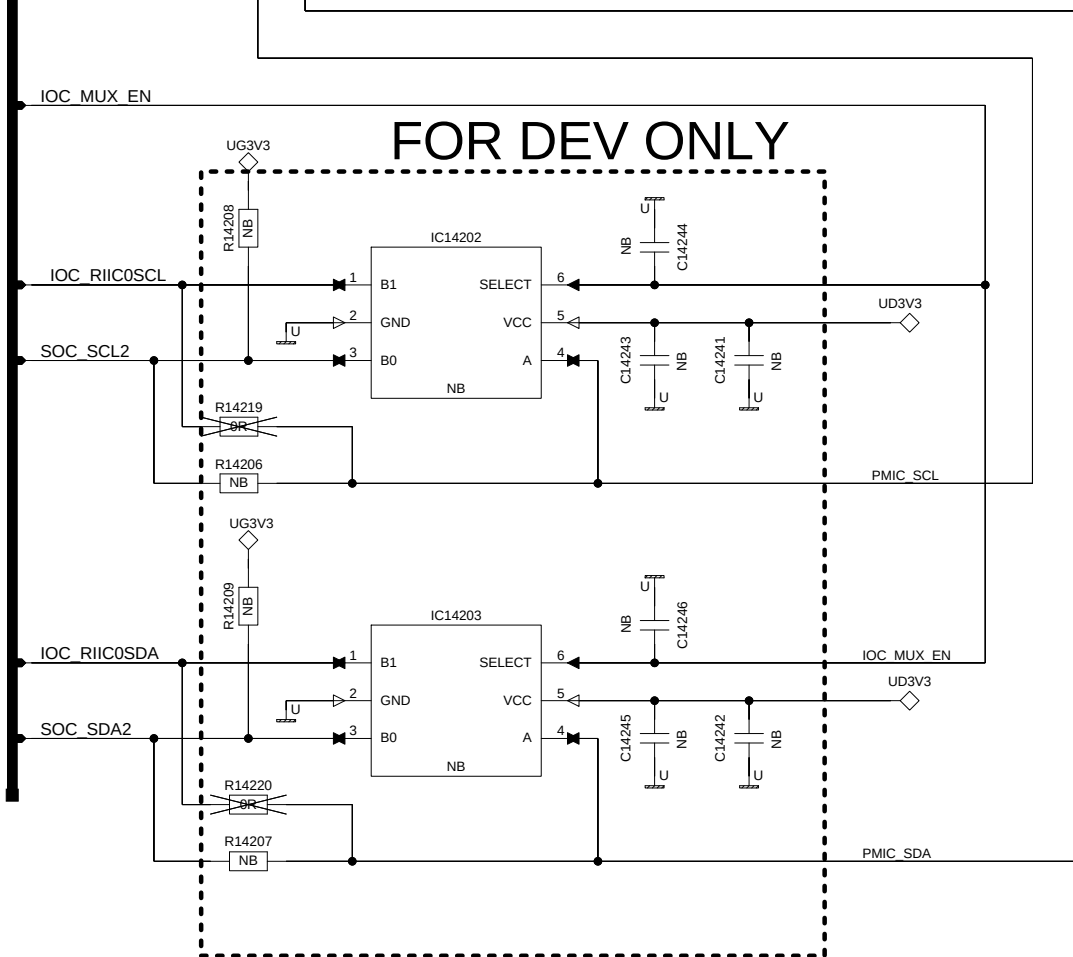
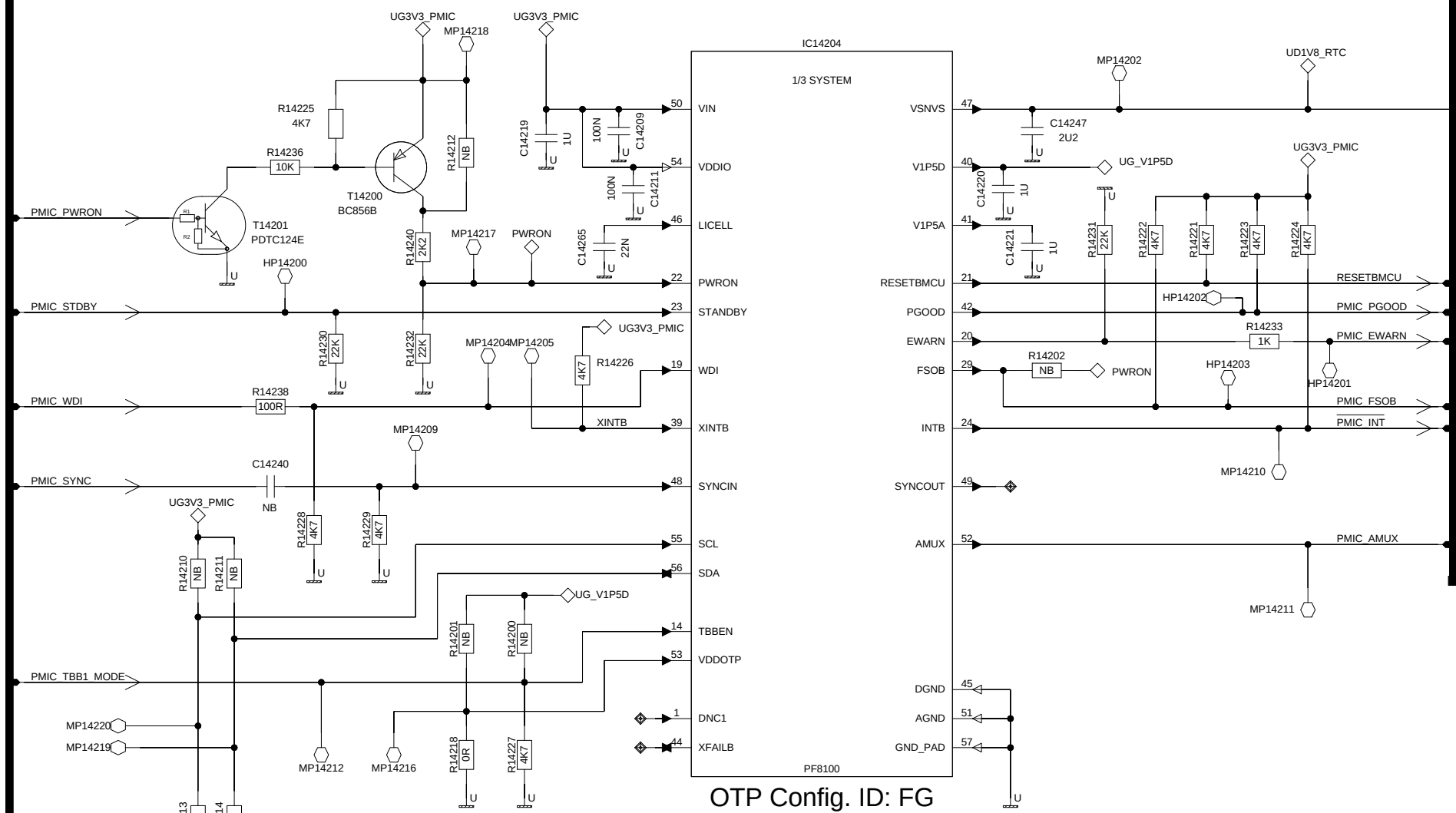
Rev.	type of change					change no.	date of change		auth.	checked
 Copyright reserved						date		name		
						created				
						checked				
schematic:						Sheet 11	Sheets of 31	Sample		
4010_IDC						Type	Variant C143		Part Number	
substitute for:						substitute by:				

MAIN CONNECTOR (5300-5399)



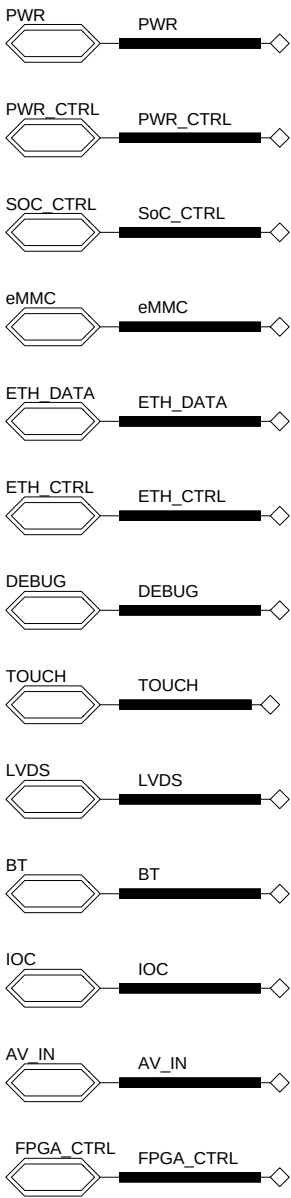
The reproduction, distribution and utilization of this document as well as the communication of its contents to others without explicit authorization is prohibited. Offenders will be held liable for the payment of damages. All rights reserved in the event of the grant of a patent, utility model or design.

Rev.	type of change		change-no.	date of change	auth. checked
		Copyright reserved	Wiring modifications reserved	date name	
				created	
				checked	
schematic			Sheet 12	Sheets of 31	Sample
4010_Main_Connector			Type	Variant	Part Number
substitute for:		substitute by:			

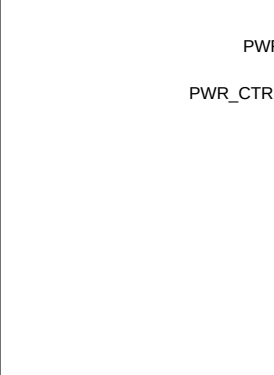


Rev.	type of change		change-no.	date of change	auth. checked
		Copyright reserved	Wiring modifications reserved		
schematic			Sheet	Sheets of	Sample
			13	31	
			Type	Variant	Part Number
substitute for:		substitute by:			

SOC (10000-10999)

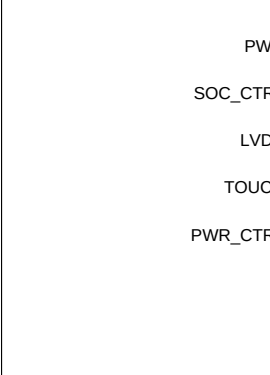


4010 SoC Power



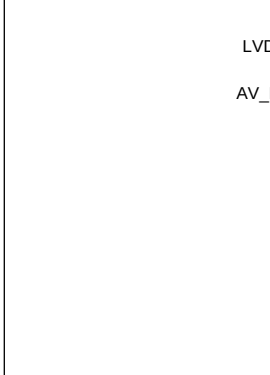
10500 - 10599

4010 SoC_CTRL



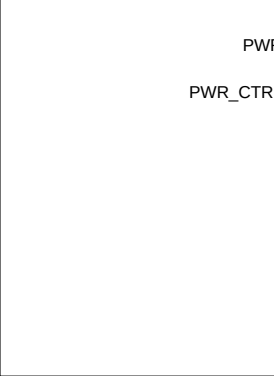
10600 - 10699

4010 SoC Video



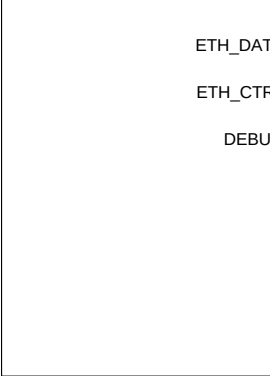
10700 - 10799

4010 SoC LPDDR4



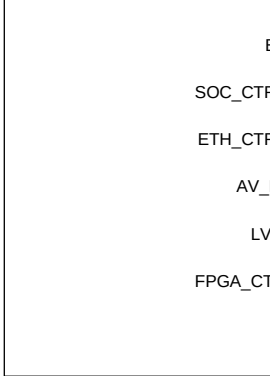
10400 - 10499

4010 SoC_ETH_PCIE_USB



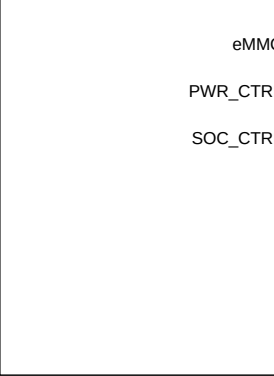
10300 - 10399

4010 SoC Audio



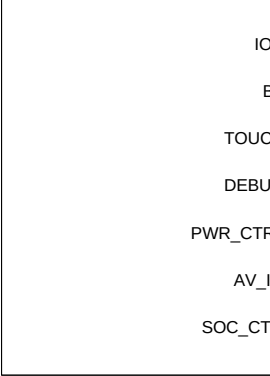
10800 - 10899

4010 SoC_eMMC_NAND_SD



10100 - 10199

4010 SoC_Serial_IF

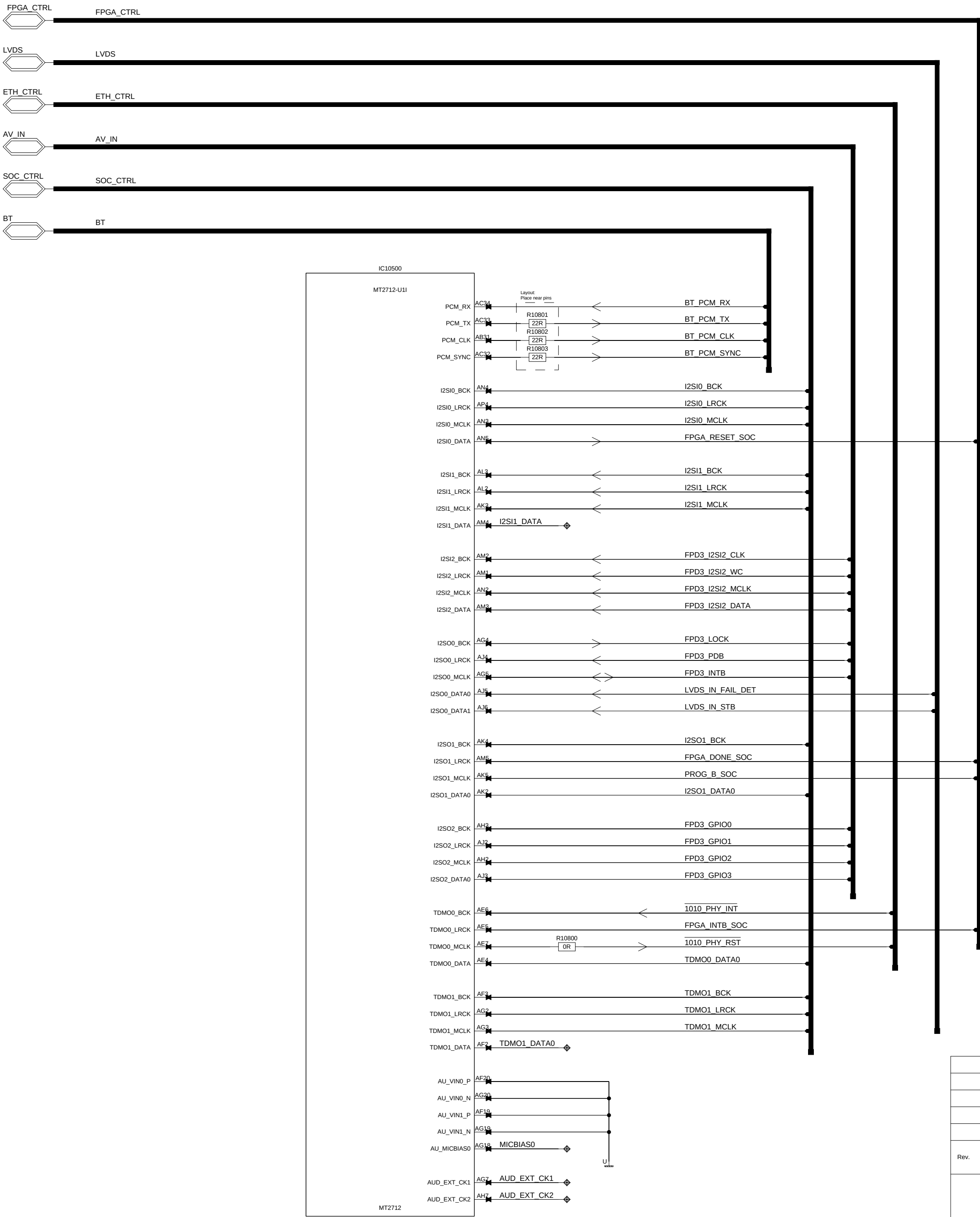


10200 - 10299

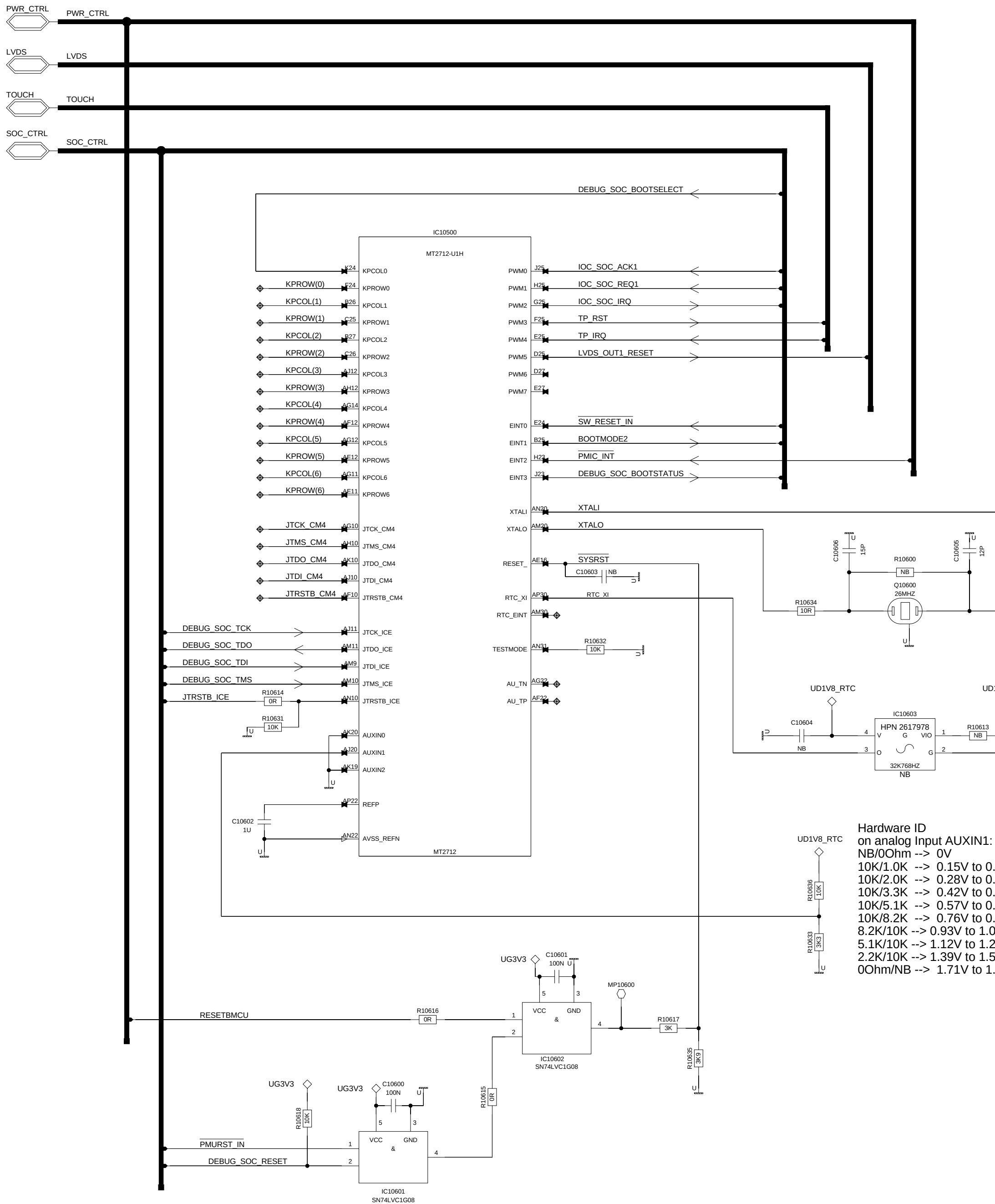
The reproduction, distribution and utilization of this document as well as the communication of its contents to others without explicit authorization is prohibited. Offenders will be held liable for the payment of damages. All rights reserved in the event of the grant of a patent, utility model or design.

Rev.	type of change		change-no.	date of change	auth. checked
		Copyright reserved	Wiring modifications reserved	date name	
				created	
				checked	
schematic			Sheet	Sheets of	Sample
4010_SoC			15	31	
substitute for:		substitute by:	Type	Variant	Part Number

SOC_AUDIO (10800-10899)



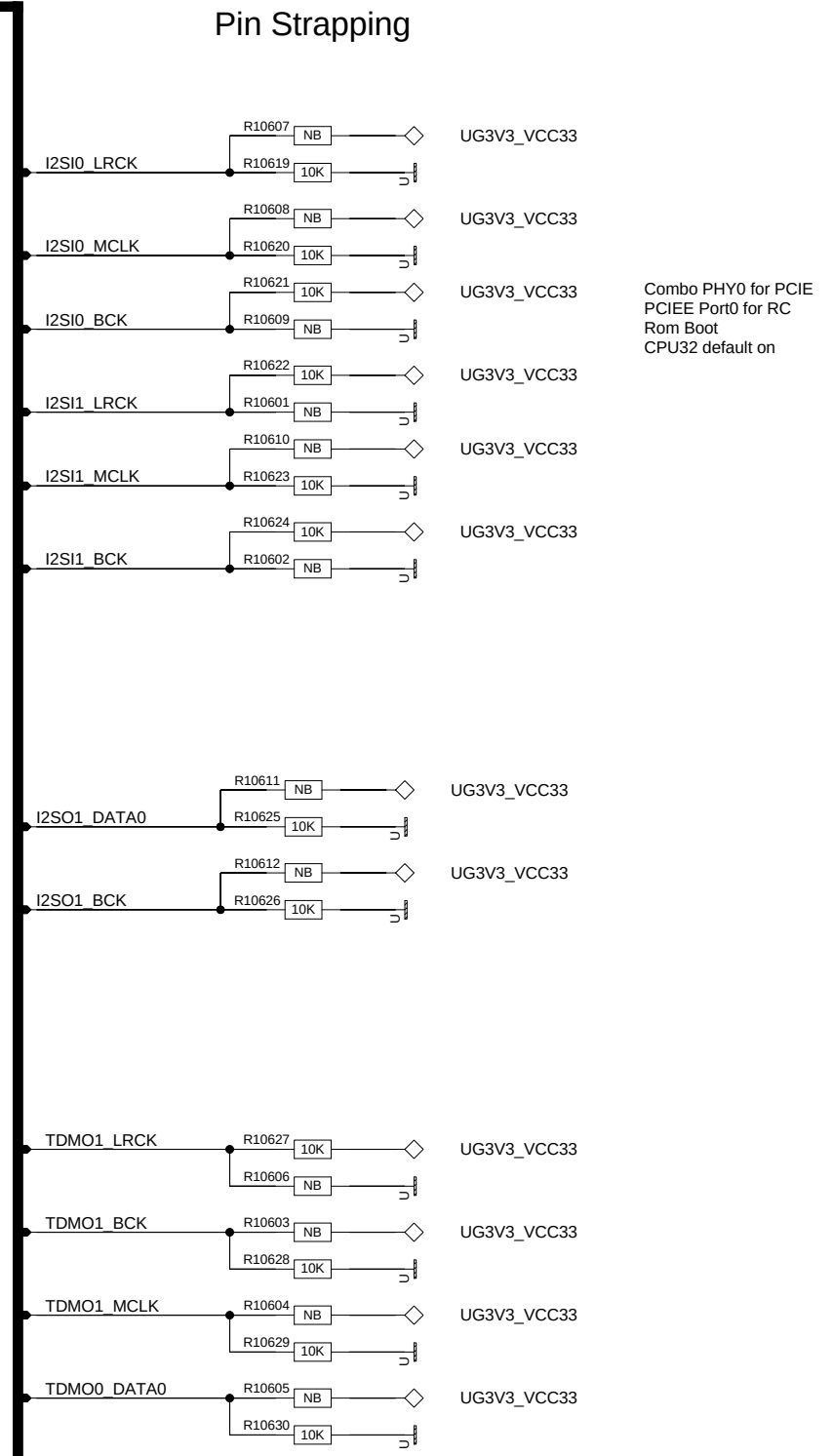
Rev.	type of change	change-no.	date of change	auth.	checked
	Copyright reserved	Wiring modifications reserved	date		name
			created		
			checked		
schematic		Sheet	Sheets of	Sample	
4010_SoC_Audio		16	31		
substitute for:		Type	Variant	Part Number	
substitute by:					



	multi-used				multi-used	
Mode	I2SiO_LRCK	I2SiO_MCLK	I2SiO_BCK	I2Si1_BCK	I2Si1_LRCLK	I2Si1_MCLK
Combo PHY0 for USB	X	X	X	X	X	1
Combo PHY0 for PCIE	X	X	X	X	X	0
Combo PHY1 for USB	X	X	X	X	1	X
Combo PHY1 for PCIE	X	X	X	X	0	X
PCIEE Port0 for RC	X	X	X	1	X	X
PCIEE Port0 for EP	X	X	X	0	X	X
PCIEE Port1 for RC	X	X	1	X	X	X
PCIEE Port1 for EP	X	X	0	X	X	X
Rom Boot	X	0	X	X	X	X
Nor-flash bypass rom boot	X	1	X	X	X	X
CPU32 default on	0	X	X	X	X	X
CPU64 default on	1	1	X	X	X	X

i2S01_DATA0 status[1]	L	L	H	H
i2S01_BCK status[0]	L	H	L	H
1	EMMC	NAND	NOR	SD
2	SD	EMMC	NAND	NOR
3	NOR	SD	EMMC	NAND
4	NAND	NOR	SD	EMMC

Trapping Mode	TDMO1_LRCK	TDMO1_BCK	TDMO1_MCLK	TDMO1_DATA0
Slowest Boot device speed configuration	1	X	X	X
Fastest Boot device speed configuration	0	X	X	X
NAND randomizer: disable	X	0	X	X
NAND randomizer: enable	X	1	X	X
NOR flash address mode: 3-byte	X	X	0	X
NOR flash address mode: 4-byte	X	X	1	X
Alternative boot loader: disable	X	X	X	0
Alternative boot loader: enable	X	X	X	1

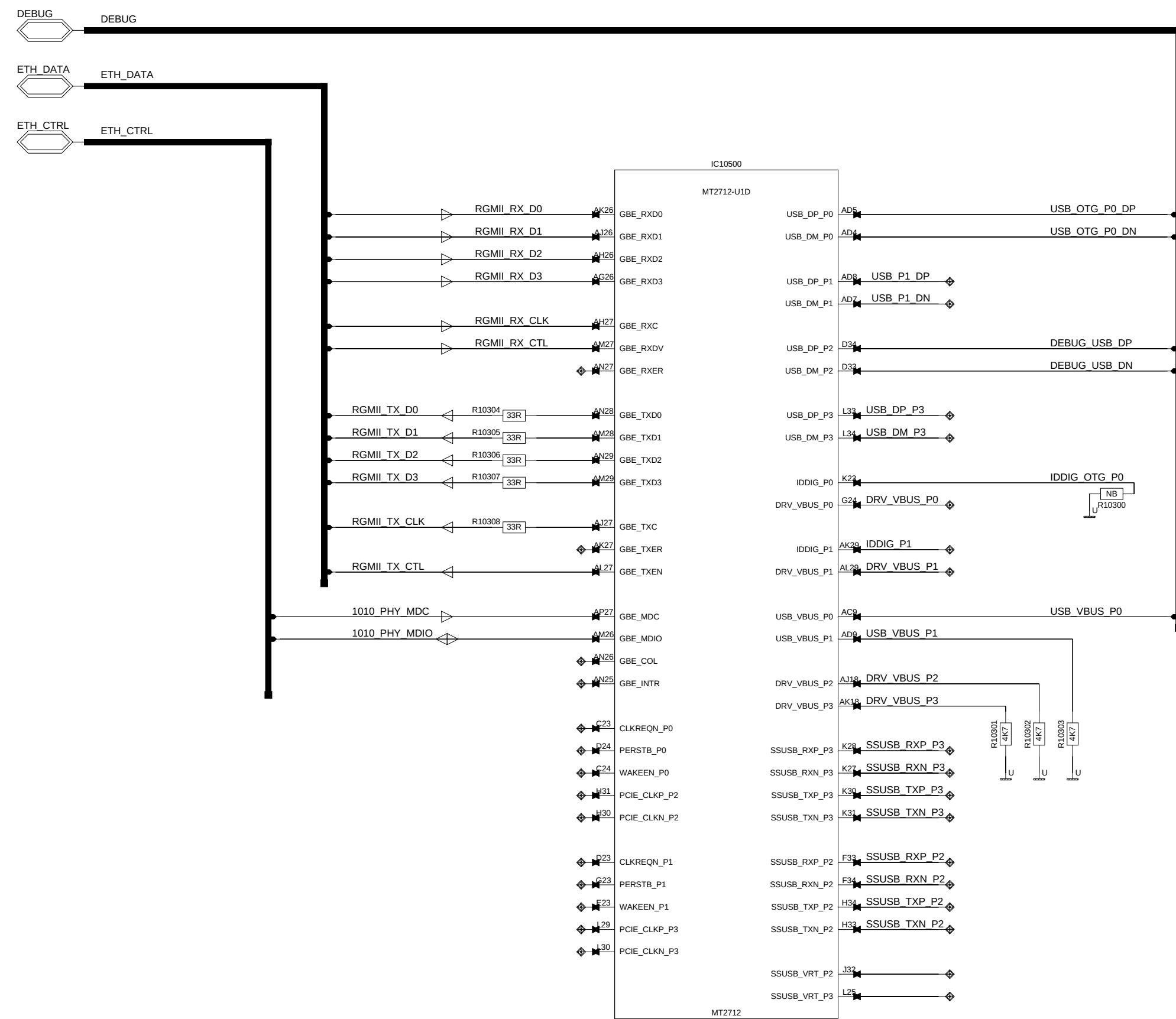


```
Hardware ID
NB/OOHm Input AUXIN1:
10K/1.0K --> 0.15V to 0.18V
10K/2.0K --> 0.28V to 0.32V
10K/3.3K --> 0.42V to 0.47V
10K/5.1K --> 0.57V to 0.65V
10K/8.2K --> 0.76V to 0.86V
8.2K/10K --> 0.93V to 1.05V
5.1K/10K --> 1.12V to 1.26V
2.2K/10K --> 1.39V to 1.55V
OOHm/NB --> 1.71V to 1.89V
```

ED1 HW
ED2 HW
ED3 HW
DV HW
PV HW
PV2 HW
PV3 HW

Rev.	type of change		change-no.	date of change	auth. checked
		Copyright reserved	Wiring modifications reserved	date name	
				created	
				checked	
schematic			Sheet 17	Sheets of	Sample
4010_SoC_CTRL			Type	Variant	Part Number
substitute for:		substitute by:			

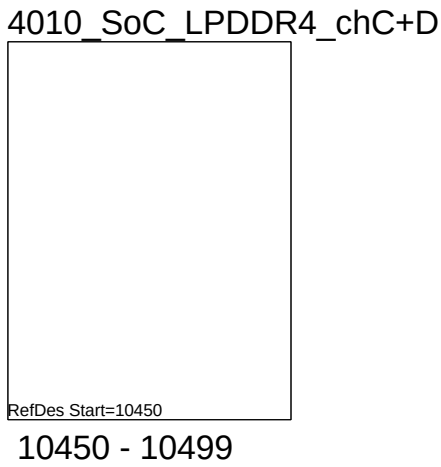
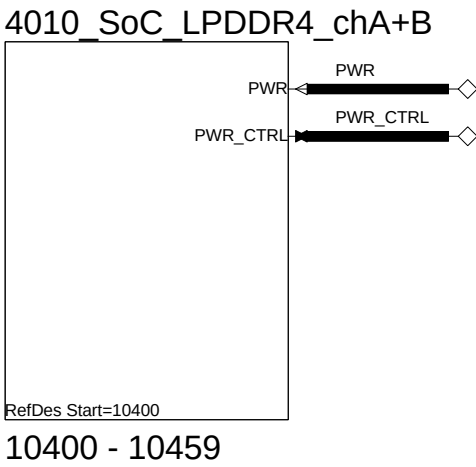
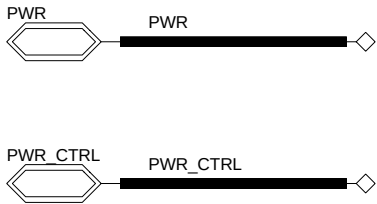
SOC_ETH_PCIE_USB (10300-10399)



Rev.	type of change		change-no.	date of change	auth. checked
		Copyright reserved	Wiring modifications reserved	date name	
				created	
				checked	
schematic			Sheet 18	Sheets of 31	Sample
4010_Soc_ETH_PCIe_USB			Type	Variant	Part Number
substitute for:		substitute by:			

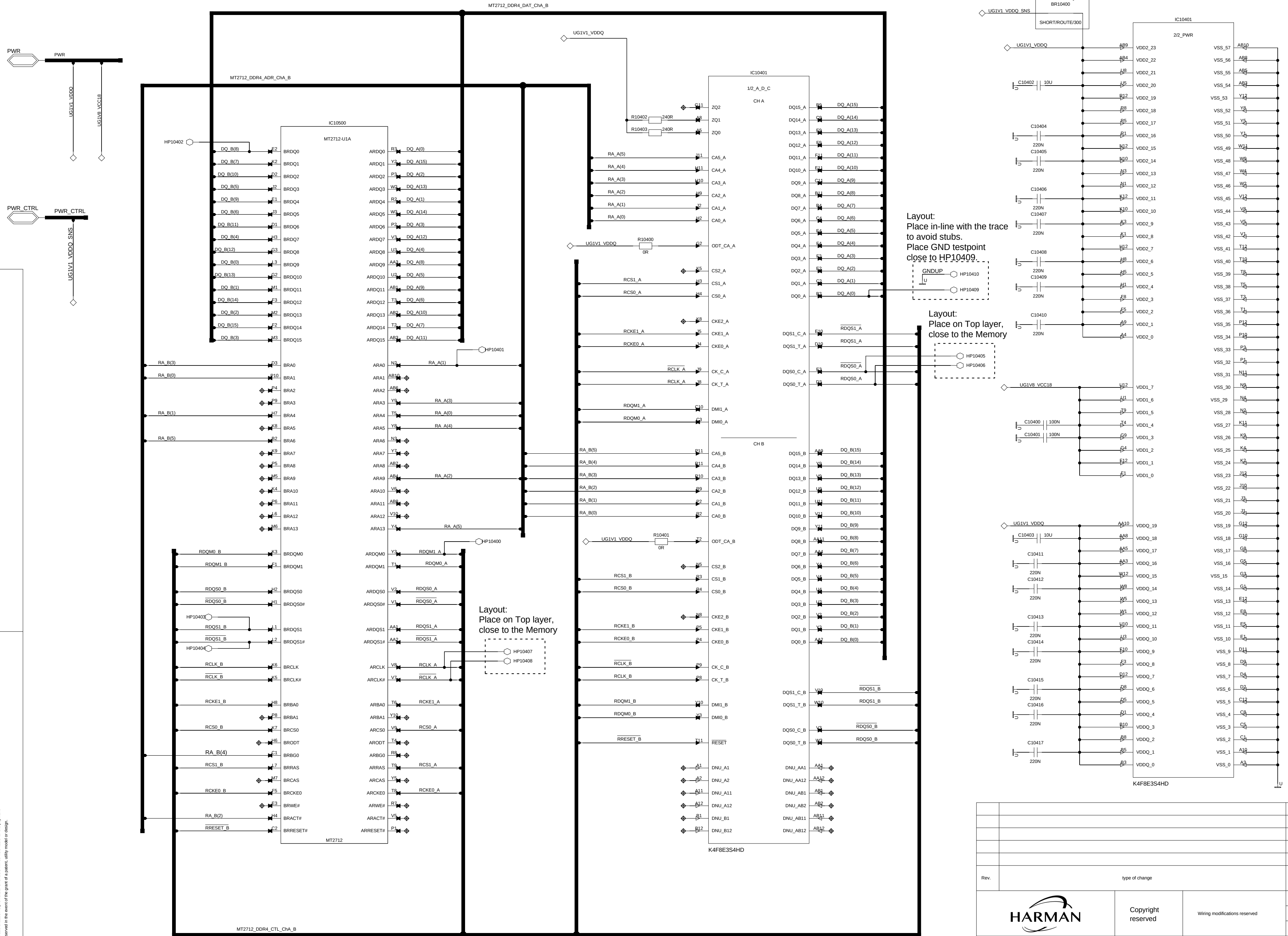
The reproduction, distribution and utilization of this document as well as the communication of its contents to others without explicit authorization is prohibited. Offenders will be held liable for the payment of damages. All rights reserved in the event of the grant of a patent, utility model or design.

SOC_LPDDR4 (10400-10499)



Rev.	type of change		change-no.	date of change	auth.	checked
		Copyright reserved	Wiring modifications reserved	date		name
				created		
				checked		
schematic			Sheet	Sheets of		Sample
4010_SoC_LPDDR4			19	31		
substitute for:			Type	Variant		Part Number

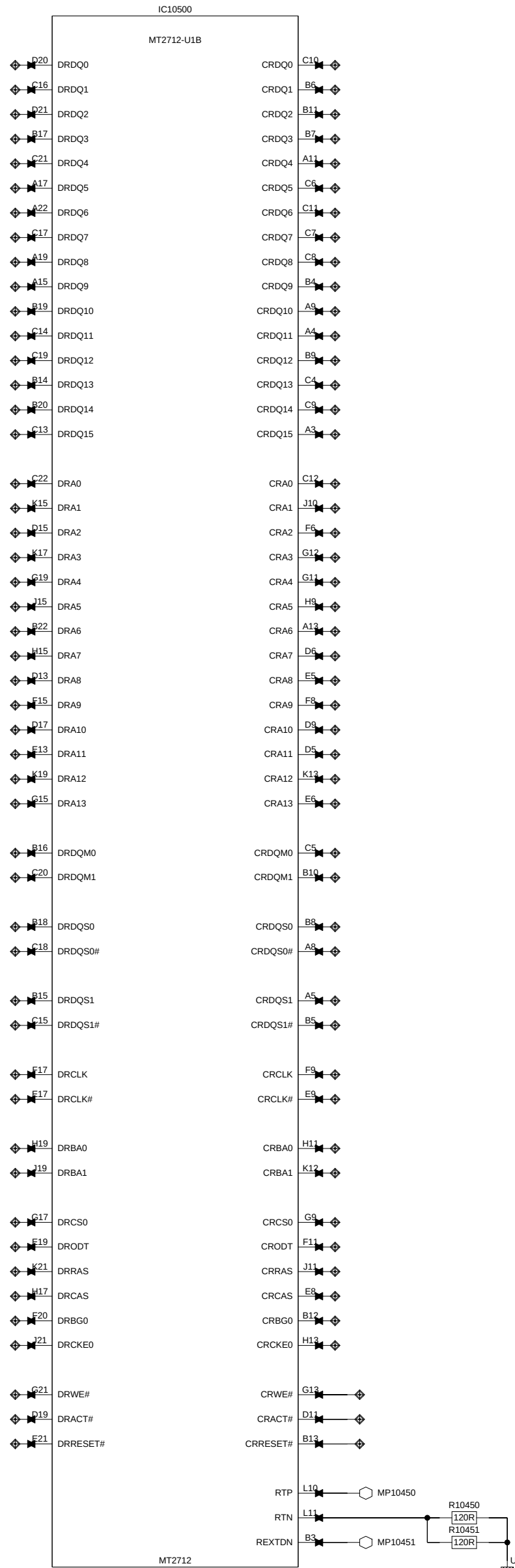
SOC_LPDDR4 (10400-10449)



The reproduction, distribution and utilization of this document as well as the communication of its contents to others without explicit authorization is prohibited. Offenders will be held liable for the payment of damages. All rights reserved in the event of the grant of a patent, utility model or design.

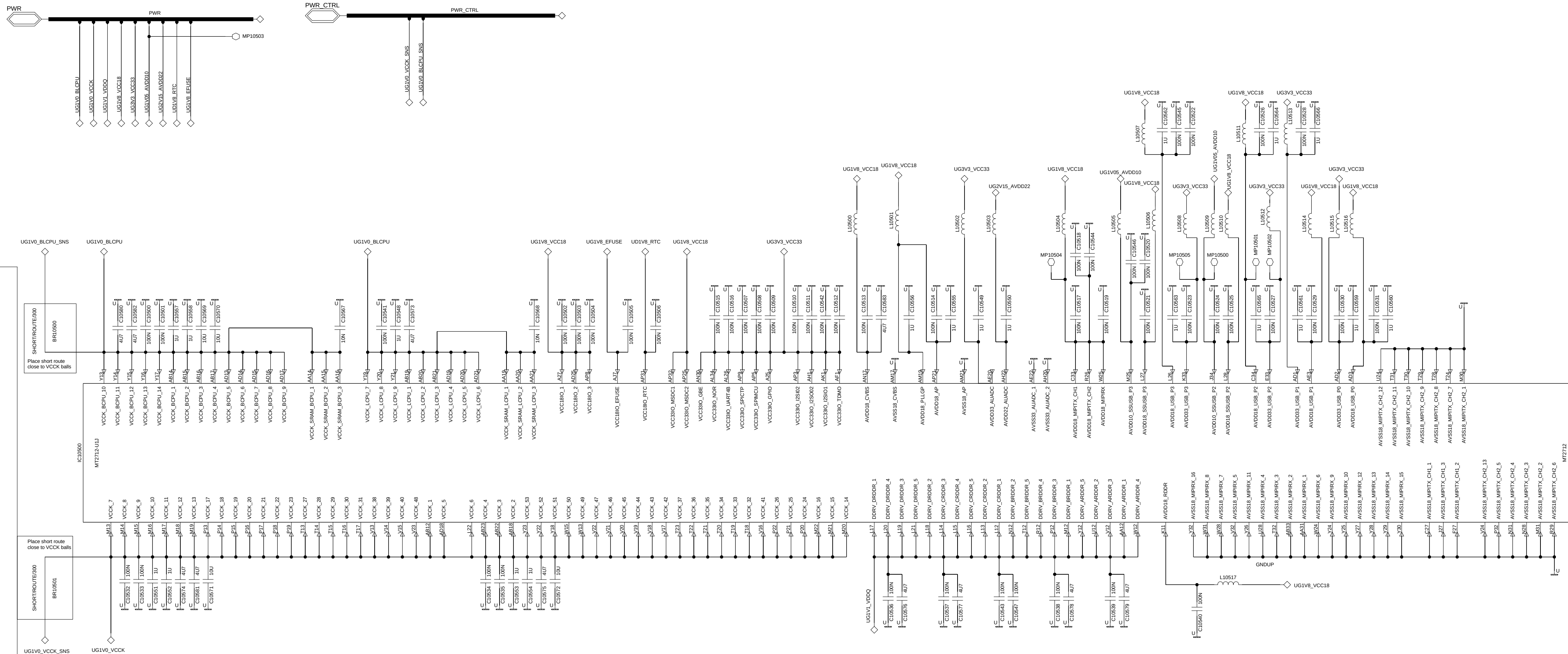
Rev.	type of change		change-no.	date of change	auth. checked
 Copyright reserved Wiring modifications reserved			date		name
			created		
			checked		
schematic			Sheet 20	Sheets of 31	Sample
4010_SoC_LPDDR4_chA+B			Type	Variant	Part Number
substitute for:		substitute by:			

SOC_LPDDR4 (10450-10499)

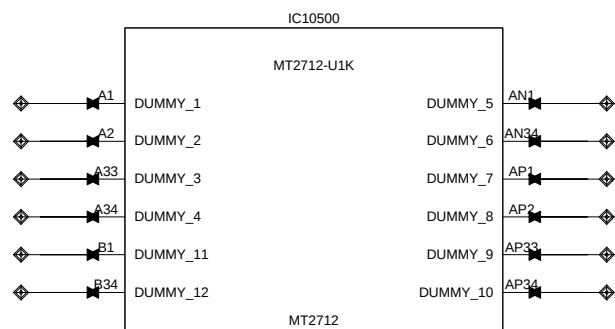


Rev.	type of change		change-no.	date of change	auth.	checked
		Copyright reserved	Wiring modifications reserved	date		name
				created		
				checked		
schematic				Sheet	Sheets of	Sample
4010_SoC_LPDDR4_chC+D				21	31	
substitute for:		substitute by:	Type	Variant	Part Number	

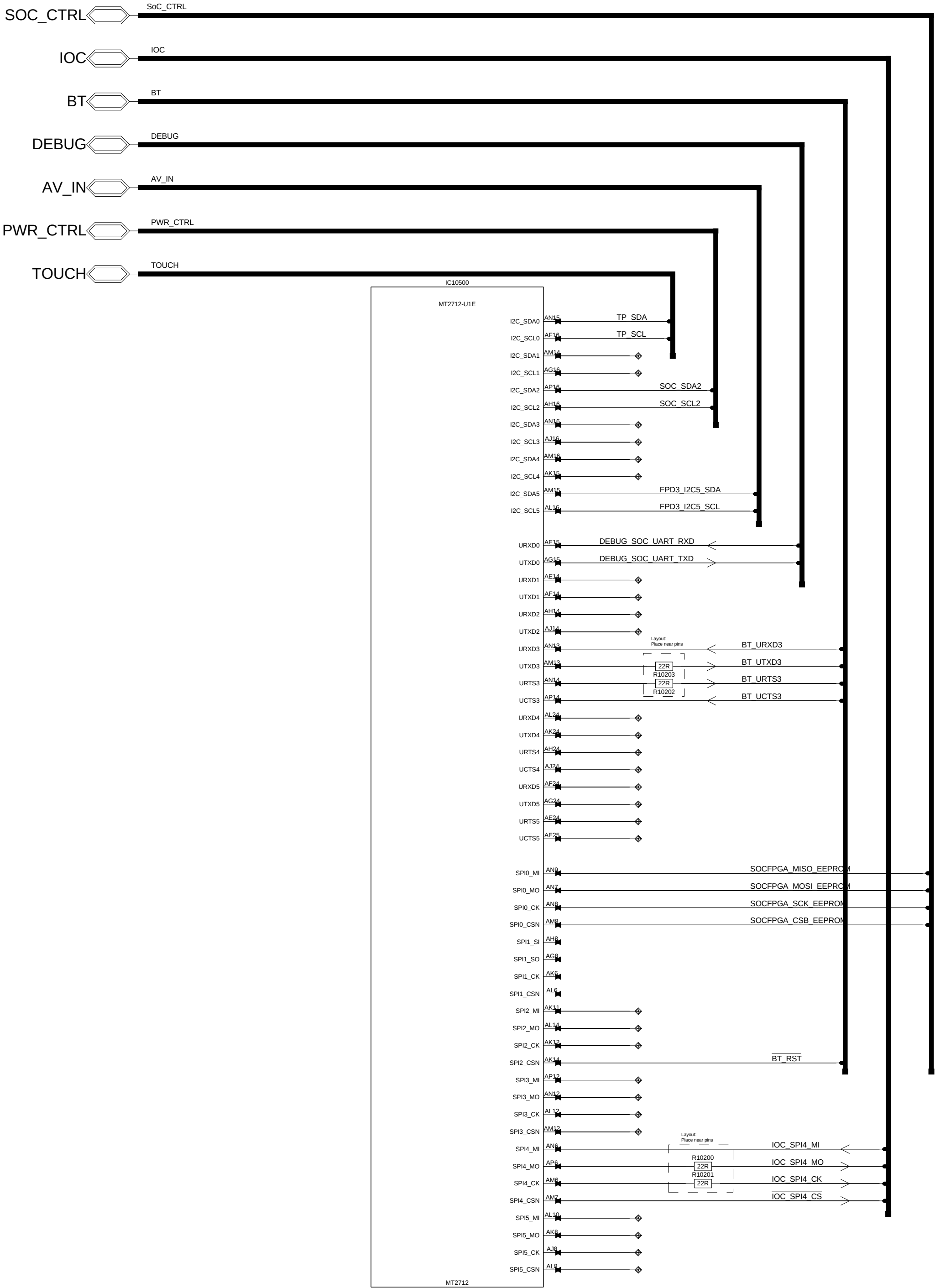
SOC_POWER (10500-10599)



4010 SoC GND

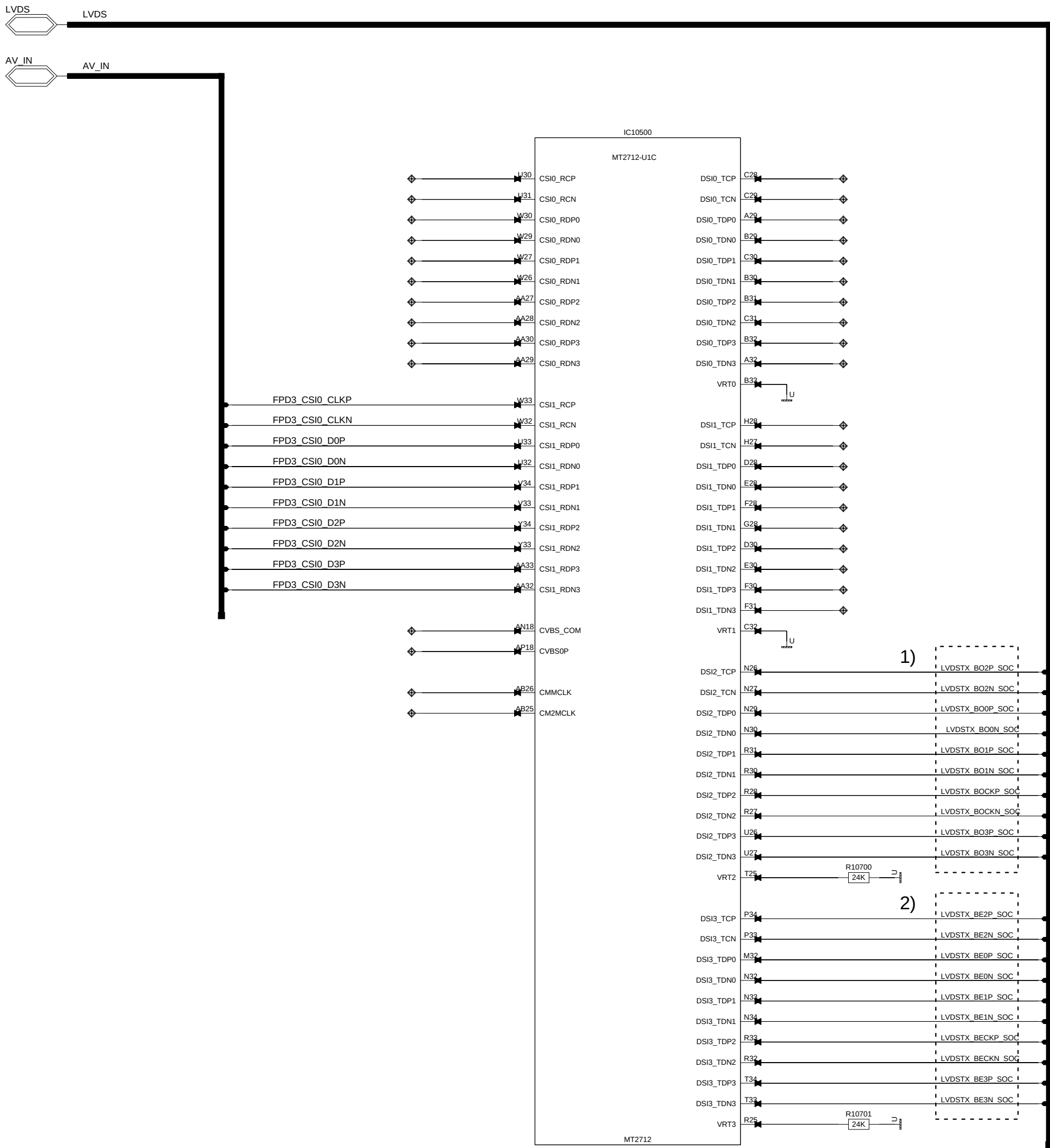


SOC_SERIAL_IF (10200-10299)



Rev.	type of change		change-no.	date of change	auth. checked
		Copyright reserved	Wiring modifications reserved	date name	
				created	
				checked	
schematic			Sheet 24	Sheets of 31	Sample
4010_SoC_Serial_IF			Type	Variant	Part Number
substitute for:		substitute by:			

SOC_VIDEO (10700-10799)

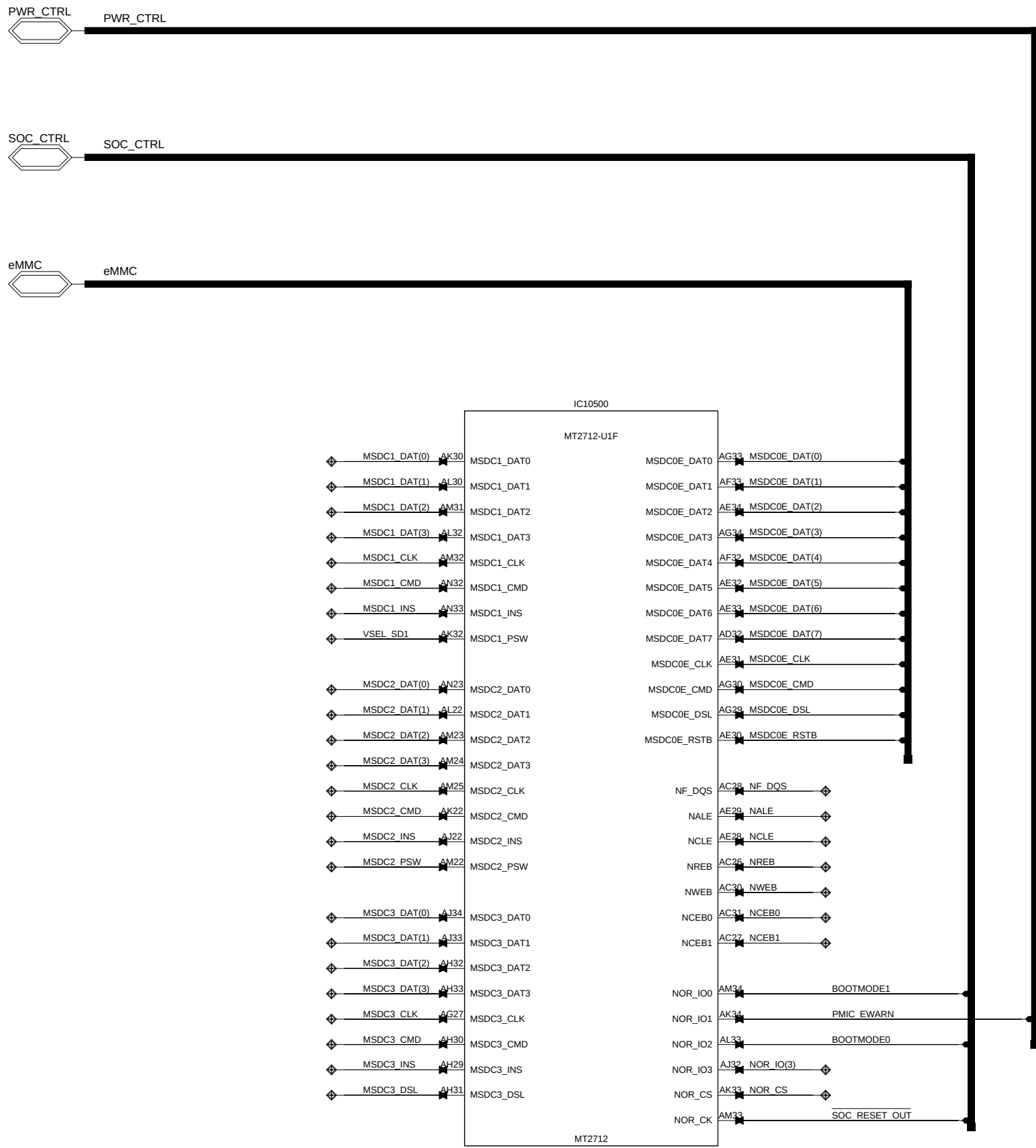


- 1) - Route as differential pair signals.
- Differential impedance: 100ohm (±10%)
- Maximum intra-pair skew: 0.127mm (5mils)
- Maximum skew from clock (BOCK) to data (BO[0-3]): 5mm

- 2) - Route as differential pair signals.
- Differential impedance: 100ohm (±10%)
- Maximum intra-pair skew: 0.127mm (5mils)
- Maximum skew from clock (BECK) to data (BE[0-3]): 5mm

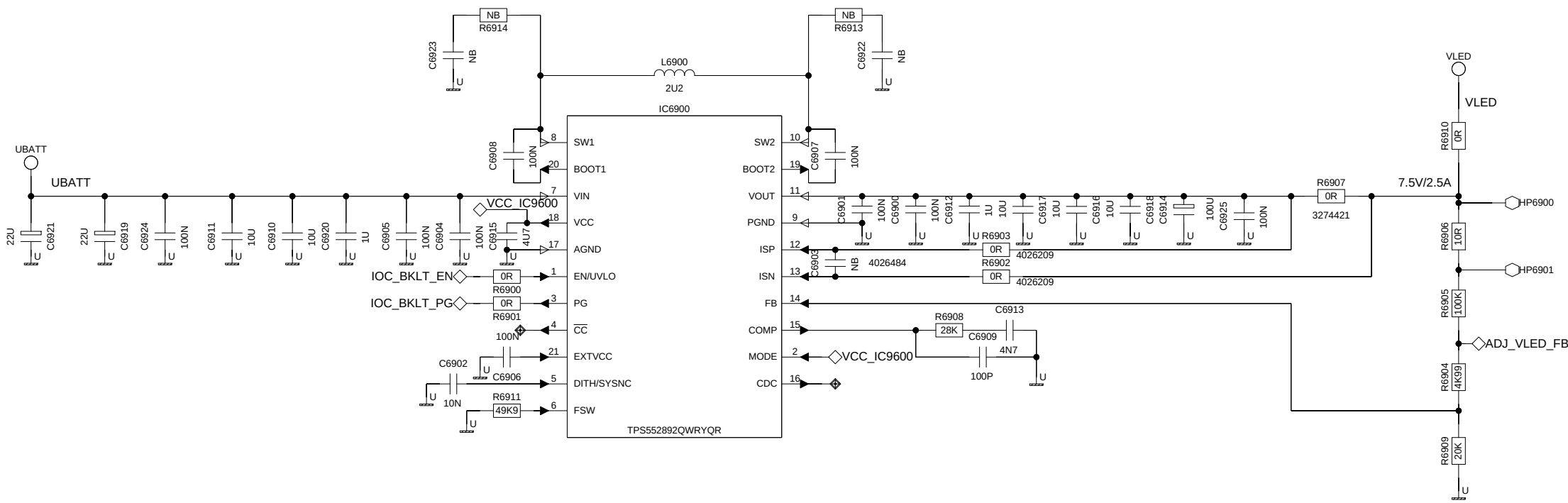
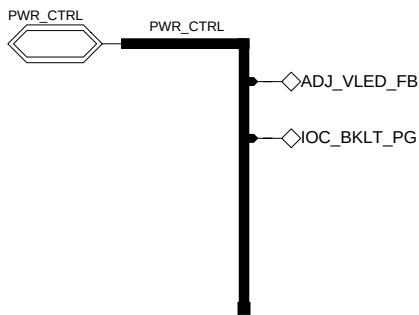
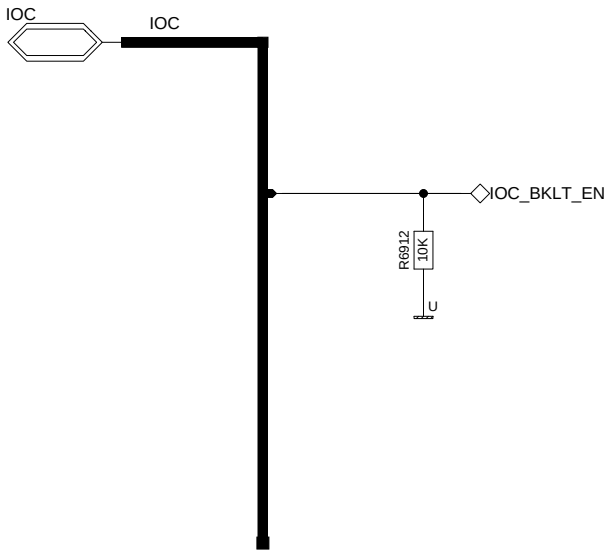
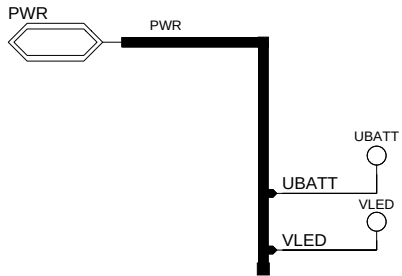
Rev.	type of change		change-no.	date of change	auth.	checked
		Copyright reserved	Wiring modifications reserved	date		name
				created		
				checked		
schematic			Sheet	Sheets of	Sample	
4010_Soc_Video			25	31		
substitute for:			Type	Variant	Part Number	
		substitute by:				

SOC_eMMC_NAND_SID (10100-10199)

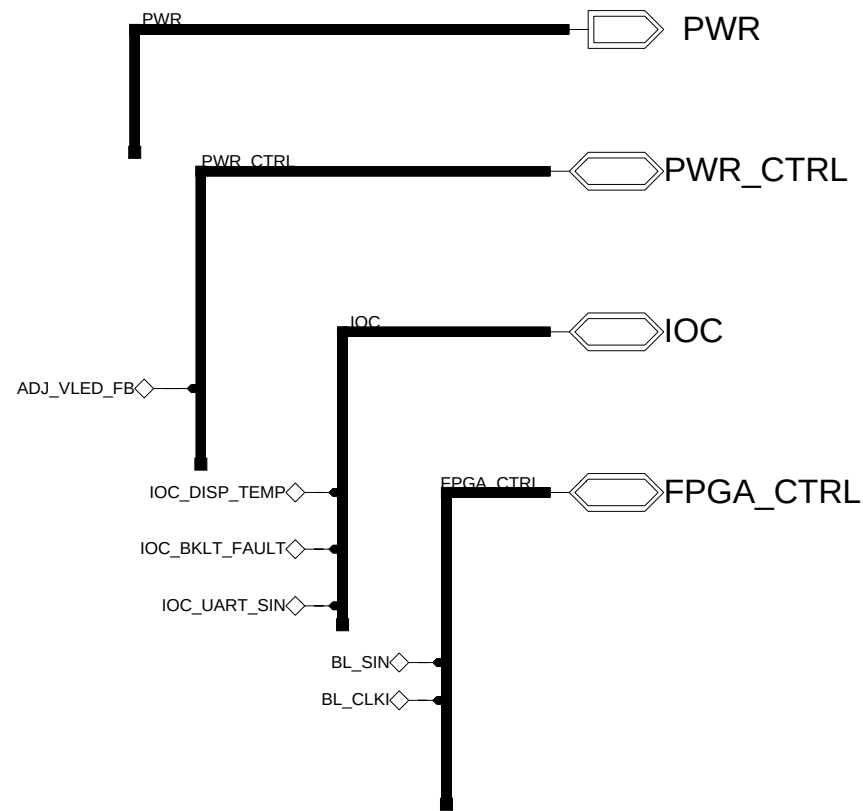


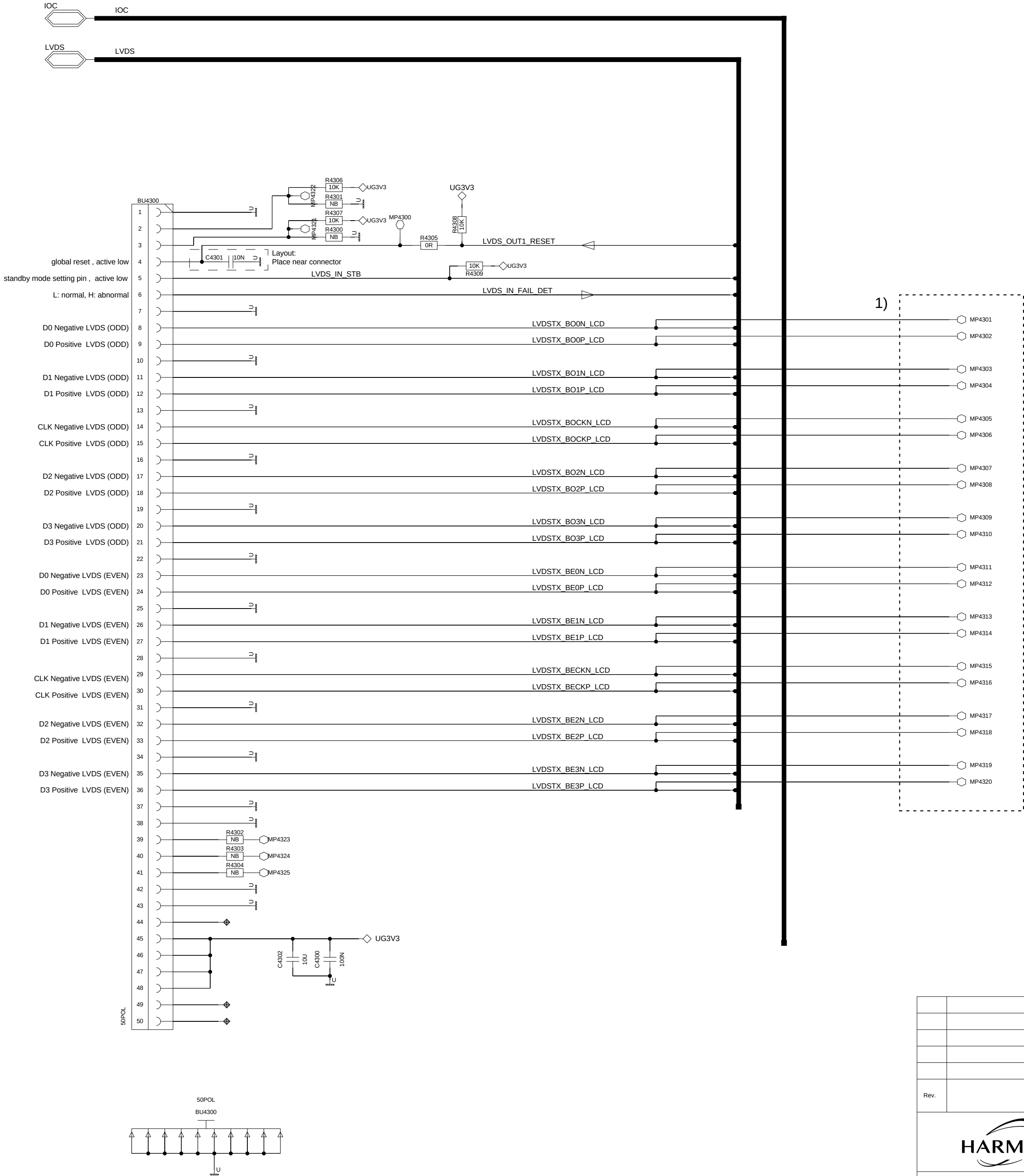
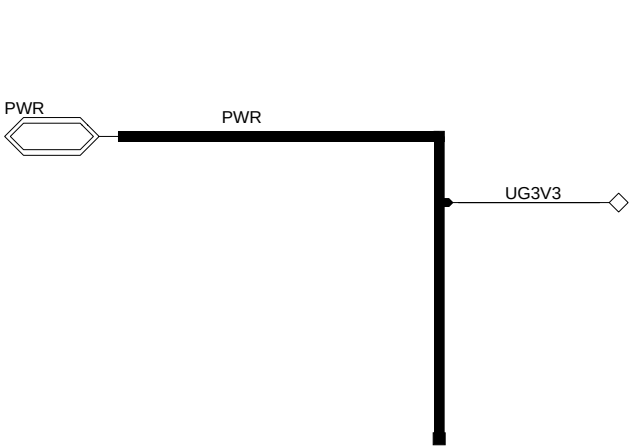
Rev.	type of change		change-no.	date of change	auth.	checked
		Copyright reserved	Wiring modifications reserved	date		name
				created		
				checked		
schematic				Sheet 26	Sheets of 31	Sample
4010_SoC_eMMC_NAND_SD				Type	Variant	Part Number
substitute for:		substitute by:				

TFT BL (6900-6999)



Rev.	type of change		change-no.	date of change	auth. checked
		Copyright reserved	Wiring modifications reserved	date name	
				created	
				checked	
schematic			Sheet 27	Sheets of 31	Sample
4010_TFT_BL			Type	Variant	Part Number
substitute for:		substitute by:		C143	

A2 26.02.12

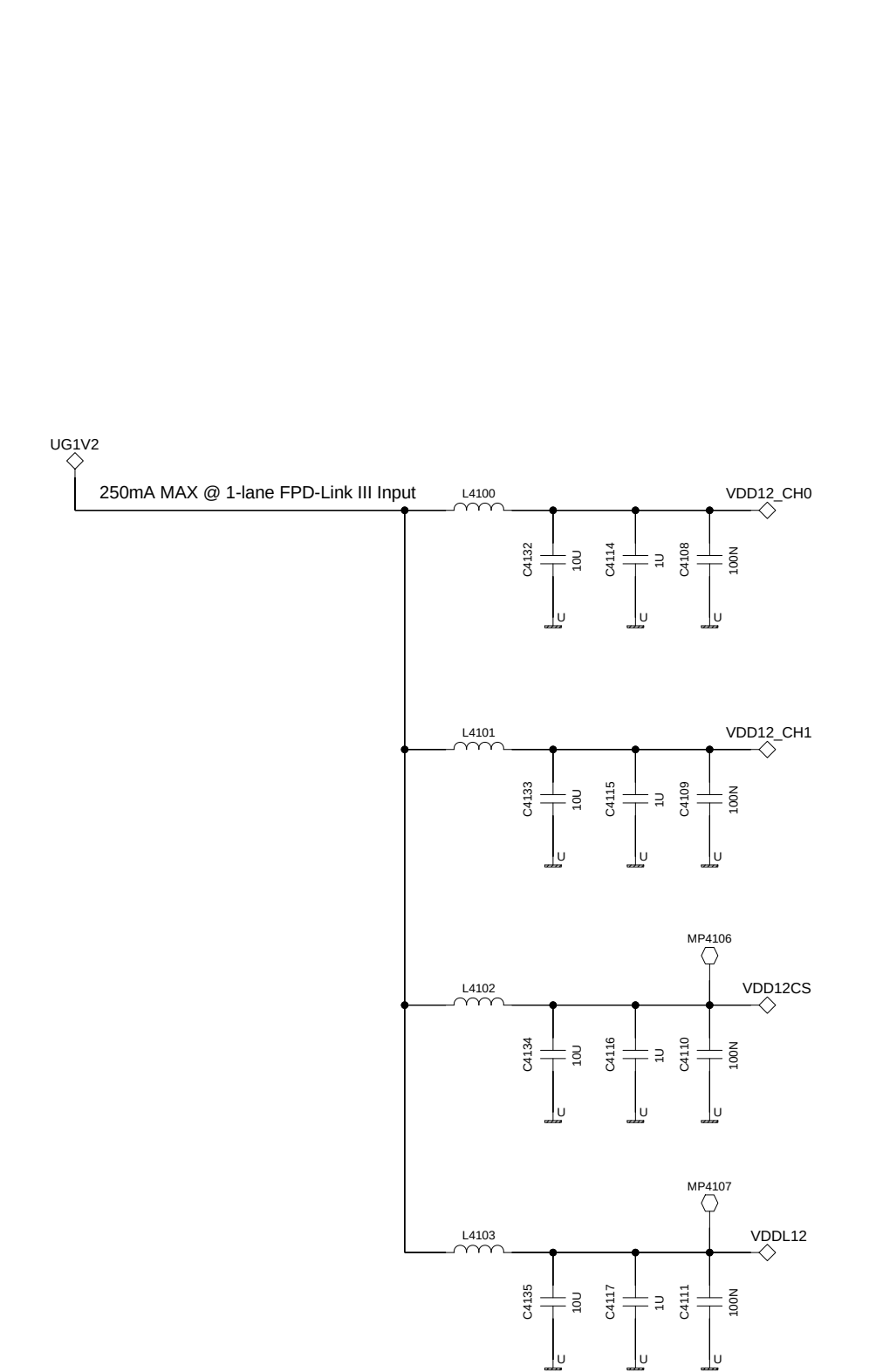


Verify pin out during layout

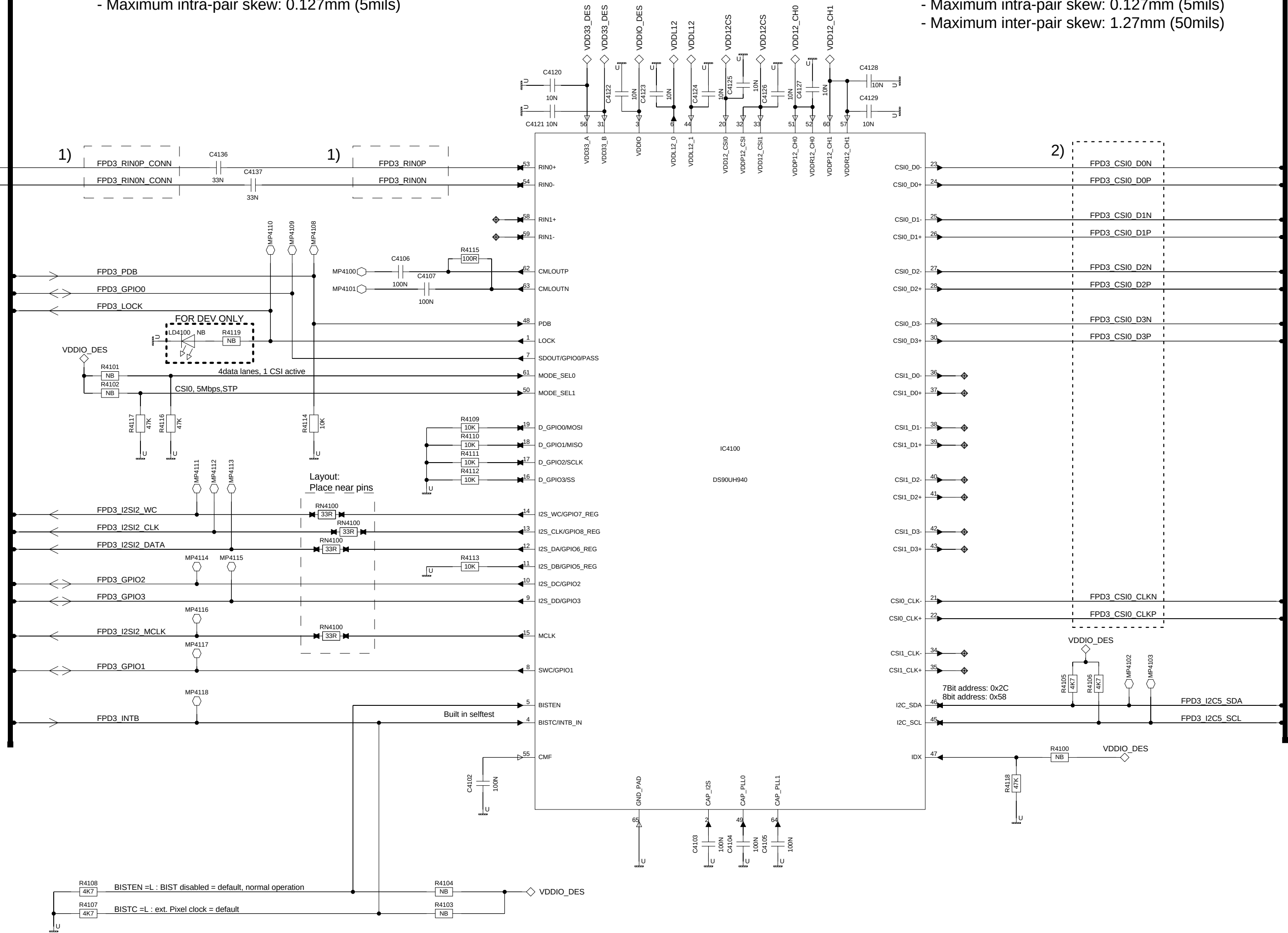
1) Testpoints should be placed as close as possible to an existing VIA

Rev.	type of change		change-no.	date of change	auth. checked
		Copyright reserved	date		name
			created		
			checked		
schematic			Sheet	Sheets of	Sample
4010_TFT_Conn			29	31	
substitute for:			Type	Variant	Part Number
				C143	
substitute by:					

The schematic diagram illustrates the power and signal connections for the AV_IN and PWR sections. The AV_IN section features a 6mA MAX @ 1-lane FPD-Link III Input, which is connected to VDDIO_DES and VDD33_DES. The PWR section features a 122mA MAX @ 1-lane FPD-Link III Input, which is connected to VDDIO_DES and VDD33_DES. The diagram includes components such as L4104, L4105, C4130, C4112, C4100, C4118, C4105, C4131, C4113, C4101, C4119, and MP4104/MP4105. A BU41100 component is also shown at the bottom left.

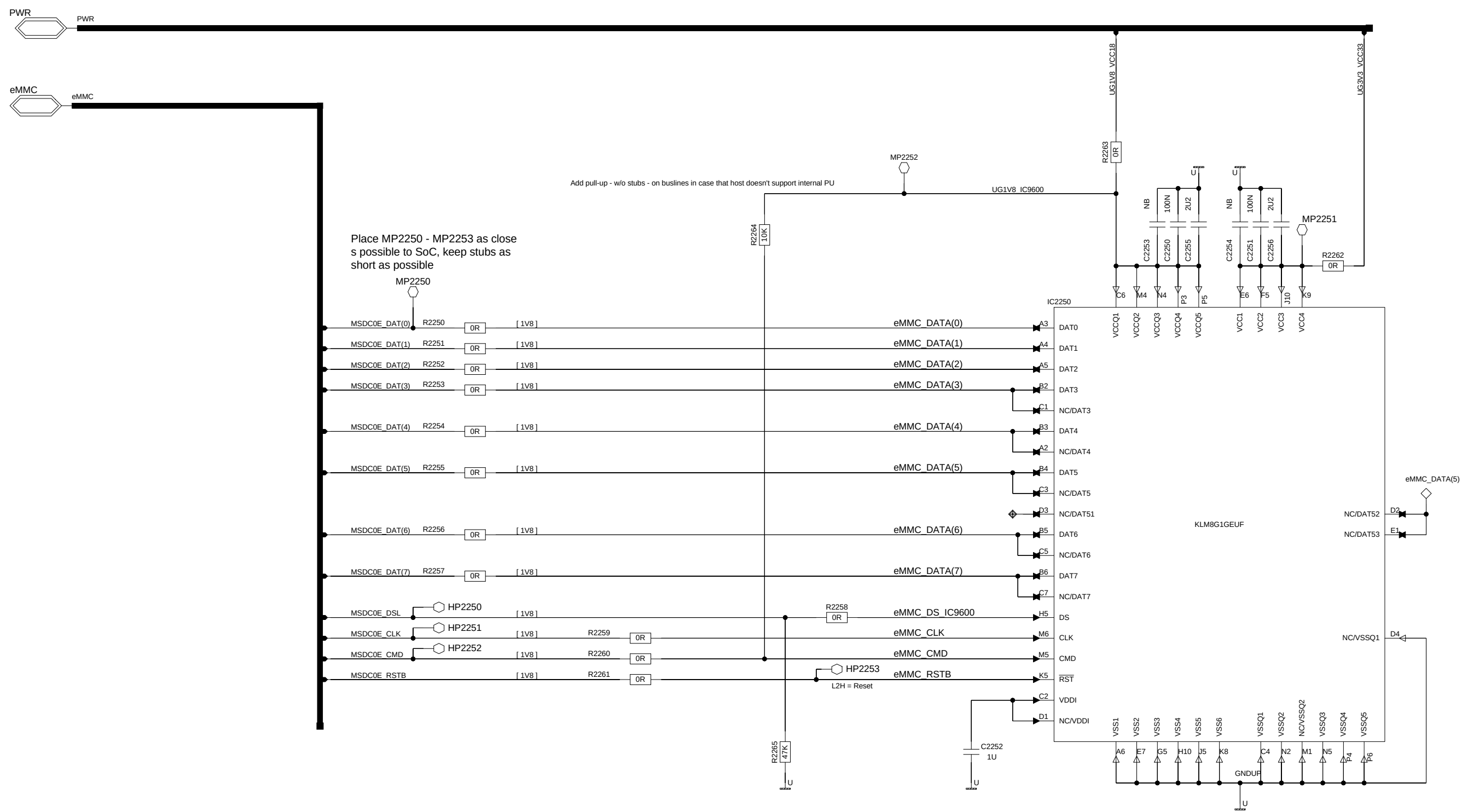


- 2) - Route as differential pair signals.
- Differential impedance: 100ohm ($\pm 10\%$)
- Maximum intra-pair skew: 0.127mm (5mils)
- Maximum inter-pair skew: 1.27mm (50mils)



Rev.	type of change		change-no.	date of change	auth. checked	
		Copyright reserved	Wiring modifications reserved	date name		
				created		
				checked		
schematic			Sheet 30	Sheets of 31	Sample	
4010_VIDEO_DES			Type	Variant	Part Number	
substitute for:		substitute by:		C143		

eMMC (2250-2299)



eMMC	Density	Part Number
Micron (-40 to 105°C)	8 GB	3881466
Micron (-40 to 85°C)	8 GB	3543617